Topic Title: Software-defined radio for lunar communications and navigation

Student Name: Haotian Yan **Student ID: z5544364**

A. **Problem statement**

Sustained lunar exploration under the LunaNet framework requires a reliable, interoperable, and autonomous Positioning, Navigation and Timing (PNT) service, yet no operational lunar navigation system or real Augmented Forward Signal (AFS) broadcasts currently exist. Compared with Earth GNSS, lunar satellites operate in highly perturbed, eccentric orbits, broadcast at lower received power, and rely on standards that are still under development, such as lunar reference frames, time scales, and ephemeris formats. These limitations prevent the testing of receiver architectures, error modelling, and PNT performance assessment using real signals. Therefore, a flexible, software-defined, end-to-end simulation environment is required to generate realistic AFS signals, model lunar-specific propagation and error effects, and evaluate user navigation performance for future lunar missions.

B. **Objective**

The objective of this project is to design and implement a complete end-to-end simulation framework for lunar navigation using the LANS Augmented Forward Signal. This includes reviewing the architecture of LunaNet and LANS, analysing lunar orbital dynamics and AFS signal characteristics, formulating pseudorange and Doppler observation models, and modelling key lunar PNT error sources. The project further aims to develop a software-defined AFS signal generator, implement a compatible receiver capable of acquisition, tracking, decoding, and PVT solution, and evaluate navigation performance, such as UERE, PDOP, and position accuracy for representative ELFO constellations and lunar south-pole user scenarios.

C. **My solution**

This project provides a full SDR-based implementation of the lunar navigation chain, beginning with high-fidelity orbit propagation using HALO-derived trajectories. Two ephemeris parameterization methods—Keplerian elements with RSW empirical accelerations and Chebyshev Cartesian polynomials are implemented to generate broadcast states. A complete AFS waveform generator is developed, incorporating dual PRN codes, BPSK modulation, LDPC/BCH-encoded navigation messages, and complex baseband I/Q synthesis. A reference receiver is implemented with acquisition, carrier-aided DLL/PLL tracking, message decoding, and least-squares PVT. Comprehensive error modelling, including DLL/PLL thermal noise, multipath, clock noise, and ephemeris-fit residuals is integrated to derive UERE and UERRE. Using this simulation environment, the system evaluates ELFO constellation coverage, geometry, and user positioning performance for a lunar south-pole user.

D. **Contributions** (at most one per line, most important first)

Modified SDR-based LANS-AFS simulator capable of multi-satellite baseband I/Q signal generation.
Implemented a fully functional AFS receiver compatible with LSIS-AFS waveform.
Constructed a simplified lunar PNT observation/error model
Evaluated ELFO constellation performance at lunar south pole using PDOP, UERE, and position-error metrics.
Implemented dual ephemeris parameterization methods and quantified accuracy vs bit-cost trade-offs.
Integrated HALO high-precision orbits to provide realistic satellite dynamics.

E. **Suggestions for future work**

Hardware-in-the-loop testing using real SDR devices (BladeRF/FE 2CH).
Evaluate multi-constellation AFS performance combining ESA LCNS, NASA LCRNS, and JAXA LNSS signals.
Develop a channel emulator more realistic with dynamics and comprehensive error

While I may have benefited from discussion with other people, I certify that this report is entirely my own work, except where appropriately documented acknowledgements are included.

Signature: Haotian

Date: 19 / 11 / 2025

Pointers

List relevant page numbers in the column on the left. Be precise and selective: Don't list all pages of your report!

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SCHOOL OF ELECTRICAL ENGINEERING
AND TELECOMMUNICATIONS

Software-defined radio for lunar communications and navigation

by

Haotian Yan

Student ID: z5544364

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Abstract

As lunar exploration shifts from short-term sorties to sustained missions under the Artemis program, there is a critical need for reliable, autonomous Positioning, Navigation, and Timing (PNT) services. The Lunar Augmented Navigation Service (LANS), proposed within the LunaNet framework, aims to address this by broadcasting the Augmented Forward Signal (AFS). However, the absence of an operational system or real AFS broadcasts, combined with the unique challenges of the lunar environment—such as highly perturbed orbital dynamics and constrained signal power—prevents the testing of receiver architectures using real-world signals.

To address this gap, this thesis designs and implements a complete end-to-end software-defined radio (SDR) simulation framework for lunar navigation. The project first establishes realistic satellite trajectories using the HALO high-precision orbit propagation tool and evaluates two broadcast ephemeris parameterization methods: Keplerian elements with empirical accelerations versus Chebyshev polynomial fitting. A fully functional AFS signal generator was developed, supporting dual-channel (Data/Pilot) BPSK modulation and LDPC/BCH channel coding. Complementing this, a compatible SDR receiver was implemented, capable of signal acquisition, carrier-aided tracking, message decoding, and Least-Squares PVT solution. Furthermore, a comprehensive error model incorporating clock stability, multipath effects, and thermal noise was constructed.

Using this simulation platform, the performance of an 8-satellite Elliptical Lunar Frozen Orbit (ELFO) constellation was evaluated for a user at the lunar south pole (Shackleton Crater). The ephemeris analysis reveals that an 11th-order Chebyshev polynomial model achieves sub-centimeter orbit fitting accuracy while maintaining reasonable broadcast bit costs. Finally, navigation solutions confirm that meter-level positioning accuracy is achievable using the AFS signal under realistic link budgets. These outcomes validate the feasibility of the LANS AFS design and provide a flexible simulation tool for assessing the navigation performance of future lunar missions.

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Prof. Andrew Dempster, for his continuous guidance, insightful feedback, and encouragement throughout the course of this project. His expertise in satellite navigation and his patient mentorship have been invaluable in shaping both the technical depth and overall direction of this thesis.

Abbreviations

AFS: Augmented Forward Signal

AWGN: Additive White Gaussian Noise

BCH: Bose–Chaudhuri–Hocquenghem (code)

BPSK: Binary Phase Shift Keying

C/N_0: Carrier-to-Noise Density Ratio

CED: Clock and Ephemeris Data

CLFO: Circular Lunar Frozen Orbit

CRC: Cyclic Redundancy Check

DFT: Discrete Fourier Transform

DLL: Delay Lock Loop

DOP: Dilution of Precision

DSSS: Direct Sequence Spread Spectrum

EIRP: Effective Isotropic Radiated Power

ELFO: Elliptical Lunar Frozen Orbit

ESA: European Space Agency

FLL: Frequency-Lock Loop

FSPL: Free-Space Path Loss

GCRS: Geocentric Celestial Reference System

GDOP: Geometric Dilution of Precision

GNSS: Global Navigation Satellite System

HALO: High-Precision Orbit Propagation Tool

ICRF: International Celestial Reference Frame

IMU: Inertial Measurement Unit

ITRF: International Terrestrial Reference Frame

JAXA: Japan Aerospace Exploration Agency

LANS: Lunar Augmented Navigation Service

LBF: Lunar Body Fixed

LCNS: Lunar Communication and Navigation System

LCRNS: Lunar Communication Relay and Navigation System

LCRS: Lunar Celestial Reference System

LDPC: Low-Density Parity Check

LIF: Lunar Inertial Frame

LNIS: LunaNet Interoperability Specification

LNSS: Lunar Navigation Satellite System

LOS: Line-of-Sight

LRT: LunaNet Reference Time

LSIS: LunaNet Signal-In-Space Interface Specification

LTC: Coordinated Lunar Time

ME: Mean Earth

MIRA: Micro Rubidium Atomic clock

NASA: National Aeronautics and Space Administration

NCO: Numerically Controlled Oscillator

NS-ELFO: North–South Elliptical Lunar Frozen Orbit

OCXO: Oven-Controlled Crystal Oscillator

OD: Orbit Determination

ODTS: Orbit Determination and Time Synchronization

PA: Principal Axis

PDOP: Position Dilution of Precision

PLL: Phase Lock Loop

PNT: Positioning, Navigation, and Timing

PRN: Pseudorandom Noise

PVT: Position, Velocity, and Time

QPSK: Quadrature Phase Shift Keying

RAAN: Right Ascension of the Ascending Node

RAFS: Rubidium Atomic Frequency Standard

RAIM: Receiver Autonomous Integrity Monitoring

RIC: Radial-Intrack-Crosstrack

RMS: Root-Mean-Square

RSW: Radial-Along-Track-Cross-Track

RTN: Radial-Transverse-Normal

RWFM: Random-Walk Frequency Modulation

S-ELFO: South Elliptical Lunar Frozen Orbit

SDR: Software Defined Radio

SISE: Signal-in-Space Error

SP: Synchronization Pattern

SRP: Solar Radiation Pressure

SWaP: Size, Weight, and Power

TDOP: Time Dilution of Precision

TOI: Time of Interval

TOT: Time of Transmission

UEE: User Equipment Error

UERE: User Equivalent Range Error

UERRE: User Equivalent Range-Rate Error

URE: User Range Error

WFM: White Frequency Modulation

WGS84: World Geodetic System 1984

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Chapter 1

1 Introduction

Sustained human and robotic exploration of the Moon will depend on reliable, autonomous Positioning, Navigation, and Timing (PNT) services rather than continuous reliance on Earth-based tracking. During the Apollo era, lunar navigation was performed almost entirely from the ground using large tracking antennas and dedicated operations teams. While this approach was adequate for short, crewed missions, it does not scale to the emerging vision of a busy lunar environment with multiple landers, orbiters, surface assets, and commercial activities operating simultaneously. Under the LunaNet framework, international partners are therefore planning a "lunar GNSS"–style infrastructure in which orbiting satellites broadcast standardized navigation signals to users in cislunar space and on the surface, enabling local, in-situ PNT solutions [2].

Within this framework, the Lunar Augmented Navigation Service (LANS) has been proposed as the functional equivalent of a GNSS for the Moon. LANS will broadcast a dedicated Augmented Forward Signal (AFS) [1] from lunar communication and navigation satellites, carrying both conventional navigation data (clock, ephemeris) and additional service information to support integrated communication and navigation. However, despite rapid progress in standardization, no operational lunar PNT constellation or AFS broadcast currently exists. Key elements such as lunar reference frames, time scales, and ephemeris formats remain under active development, and early missions will likely operate with a small number of satellites and limited ground tracking support. These factors make it difficult to test receiver architectures, observation models, and end-to-end performance using real signals.

Therefore, there is a clear need for a flexible, software-defined radio (SDR) testbed that can emulate future LANS signals and lunar environments before any operational system is deployed. Such a testbed must generate realistic AFS baseband waveforms from multiple lunar satellites, model lunar-specific propagation and error sources, and feed these signals into a reference receiver chain capable of acquisition, tracking, decoding, and PVT (position, velocity, and time) solution. It must also support representative constellation configurations—such as Elliptical Lunar Frozen Orbits (ELFOs) optimized for south-polar coverage—and provide tools to analyse geometry (PDOP), User Equivalent Range Error (UERE), and resulting position accuracy in realistic scenarios.

This thesis addresses these needs by developing an end-to-end lunar navigation simulation framework centred on the LANS Augmented Forward Signal. High-fidelity orbits, derived from dedicated lunar dynamics tools, are used to represent navigation satellites in ELFO constellations. Two complementary ephemeris parameterisation strategies are implemented—Keplerian elements with empirical accelerations in the Radial/Along-track/Cross-track (RSW) frame, and Chebyshev Cartesian polynomials—to investigate trade-offs between accuracy, message size, and user-side complexity. On top of these orbits, a complete AFS waveform generator is built, including dual PRN spreading codes, BPSK modulation, LDPC/BCH-encoded navigation messages, and complex baseband I/Q synthesis compatible with SDR front-ends.

On the user side, a software-defined AFS receiver is implemented by extending an open-source SDR framework to support the LANS signal structure [9]. The receiver performs DFT-based acquisition, carrier-aided DLL/PLL tracking on data and pilot channels, navigation message decoding, and least-squares PVT computation. A lunar PNT error model is integrated into this chain, combining contributions from satellite clock noise, broadcast ephemeris residuals, receiver thermal noise, and multipath into UERE and User Equivalent Range-Rate Error (UERRE) metrics. By running the complete loop—from orbit propagation through signal generation to receiver processing and error analysis—the framework enables a quantitative evaluation of how an ELFO-based AFS constellation would perform for a user located near the lunar south pole (e.g., Shackleton Crater).

In summary, this work provides a software-only, extendable platform for exploring the design space of lunar navigation systems under the LunaNet/LANS concept. It allows different constellation geometries, ephemeris representations, clock qualities, and signal-to-noise conditions to be tested without hardware, and yields insight into the relationship between system-level design choices (orbits, signals, clocks) and user-level performance (UERE, PDOP, position accuracy).

The objective of this thesis is to develop an end-to-end simulation and evaluation framework for the Lunar Augmented Navigation Service (LANS) Augmented Forward Signal (AFS). The work focuses on modelling lunar orbits, generating AFS baseband signals, implementing a compatible SDR receiver, and assessing navigation performance for south-polar users.

The specific goals are to:

1. Model ELFO-based lunar navigation constellations and analyse their geometric visibility for south-pole coverage.
2. Implement a software-defined AFS signal generator that produces multi-satellite baseband I/Q waveforms following the LSIS-AFS specification.
3. Extend an SDR receiver to process AFS signals, including acquisition, tracking, data decoding, and PVT computation.
4. Build a lunar PNT error model to characterise UERE contributions from clocks, ephemeris, tracking noise, and multipath.
5. Evaluate positioning performance under different constellation geometries, signal conditions, and ephemeris representations.

Chapter 2

2 Background

2.1 Evolution of Lunar Communication and Navigation Systems

The challenge of navigating and communicating on the Moon has driven significant evolution in systems and concepts from the Apollo era to today. During the Apollo program, lunar spacecraft had no dedicated navigation satellites; instead, they relied on Earth-based tracking networks (e.g., NASA's Deep Space Network) to determine their position and velocity. This ground-based approach proved effective for short-term missions but was inherently limited by Earth line-of-sight constraints and the availability of tracking stations [30]. As lunar exploration shifts toward sustained operations (e.g., Artemis and commercial missions), the prospect of numerous simultaneous landers, orbiters, and surface assets has made Earth-centric navigation support hard. So there is a growing requirement that autonomous, in-situ Positioning, Navigation, and Timing (PNT) services will be required to ensure reliable navigation on and around the Moon.

2.2 Overview of LunaNet and LANS Architecture

In response to these needs, international space agencies are developing LunaNet, an integrated "network of networks" framework intended to provide robust lunar communications and navigation services. LunaNet [2] is designed as an interoperable architecture where multiple actors (NASA, ESA, JAXA, and others) can each contribute nodes or constellations that refer to common standards, enabling a shared lunar information infrastructure [18]. Within this framework, the Lunar Augmented Navigation Service (LANS) has emerged as the functional equivalent of a Global Navigation Satellite System (GNSS) for the Moon. LANS is designed to offer positioning and timing services to users in lunar orbit and on the surface by broadcasting navigation signals from orbiting satellites, so it can replace Earth-based tracking.

The cornerstone of LANS is a new signal standard known as the Augmented Forward Signal (AFS) – a dedicated lunar navigation signal that carries both conventional GNSS-like navigation data and additional service information to users in the lunar vicinity [1]. Each LunaNet service provider node (e.g., a navigation satellite or lunar relay) will transmit the AFS in the S-band frequency (around 2492 MHz), with coverage spanning the Moon's service volume [22]. This signal is termed "forward" because it is broadcast from the infrastructure to users (similar to how GNSS satellites transmit to receivers on Earth), and "augmented" because it transmits more than just basic orbit and time data, including networking information to facilitate communication links and network access for lunar missions. In a word, AFS serves a dual role: it delivers the necessary ranging signals and orbital data for navigation, while also providing a channel for disseminating lunar network information (such as upcoming communication contact schedules or alerts) as part of an integrated lunar navigation and communication framework [8].

2.3 Key Challenges in Lunar PNT

Lunar Positioning, Navigation, and Timing (PNT) systems face a range of unique challenges distinct from terrestrial Global Navigation Satellite Systems (GNSS) due to the Moon's dynamic orbital environment, limited infrastructure, and early-stage standardization under the LunaNet framework [22]. Unlike Earth's stable, circular GNSS constellations, lunar satellites are subjected to strong third-body perturbations from Earth and the Sun, making most circular orbits unstable and requiring frequent station-keeping. For this reason, Elliptical Lunar Frozen Orbits (ELFOs) are considered prime candidates for navigation satellites, as they maintain long-term stability and continuous visibility over key regions such as the lunar South Pole. Accurate orbit determination and time synchronization (ODTS) [21] are also crucial since errors directly propagate into the User Equivalent Range Error (UERE) [19]. Moreover, conventional Keplerian-based ephemeris models are inadequate for describing the perturbed, elliptical trajectories of lunar satellites, prompting the need for more flexible parameterizations such as Chebyshev polynomials or numerical state vectors that balance accuracy and message efficiency [3]. The establishment of lunar reference frames and time scales represents another core challenge: whereas GNSS is associated with Earth's geoid and GPS time, lunar PNT requires a consistent geodetic frame and a dedicated temporal standard [12]. Two candidate frames—the Mean Earth (ME) and Principal Axis (PA) frames—are currently under evaluation, and LunaNet proposes the LunaNet Reference Time (LRT) as a timing standard aligned with the Augmented Forward Signal (AFS) and linked to a future Coordinated Lunar Time (LTC). Ensuring operational integrity further complicates the design; lunar systems must provide reliable fault detection and performance assurance under sparse monitoring coverage and limited satellite geometry. While Receiver Autonomous Integrity Monitoring (RAIM) and cross-satellite consistency checks can mitigate some anomalies, their effectiveness is constrained by the small number of satellites initially available. Receiver algorithms must adapt to these conditions, as traditional snapshot solutions requiring four or more signals are often infeasible. Lunar rovers and landers will instead depend on hybrid approaches combining AFS observations with onboard sensors such as inertial measurement units (IMUs), cameras, or altimeters to achieve continuous, robust localization.

2.4 Developing lunar navigation systems

In recent years, NASA, ESA, and JAXA have been working closely to establish the foundations of a cooperative lunar navigation network. Each agency is developing satellite systems capable of broadcasting the Augmented Forward Signal (AFS), which will serve as the backbone of the Lunar Augmented Navigation Service (LANS). Within this broader framework, NASA's Lunar Communication Relay and Navigation System (LCRNS), ESA's Lunar Communication and Navigation System (LCNS)—developed under the Moonlight program—and JAXA's Lunar Navigation Satellite System (LNSS) are positioned to become the first generation of AFS-capable service providers operating within the emerging LunaNet architecture [22].

Because these systems are being developed by different agencies with distinct technical heritages, maintaining cross-compatibility has become a central design priority. To that end, the three organizations are jointly drafting the LunaNet Interoperability Specification (LNIS) [2], a common standard intended to unify service interfaces, message formats, and time reference definitions. The goal is to ensure that individual systems—although independently managed—can operate seamlessly as components of one coherent lunar network. The latest release, LNIS Version 5, specifies the essential technical parameters and operational procedures required to achieve consistent, interoperable Positioning, Navigation, and Timing (PNT) performance across all LunaNet-compliant service providers [18].

Chapter 3

3 Literature review

3.1 Orbit Design and Constellation Optimization

Recent research on lunar navigation constellation design has focused on the trade-offs between orbit geometry, coverage, position dilution of precision (PDOP), and system robustness across various frozen-orbit families [6]. For the South Elliptical Lunar Frozen Orbit (S-ELFO), coverage increases with both semi-major axis and inclination. Simulations show that achieving ≥99% four-satellite coverage over the lunar south-pole region requires at least 12 satellites (e.g., 12/3/1 or 12/4/1 Walker patterns) when $a \leq 16{,}000$km, while smaller constellations (6–8 satellites) can still achieve 90–95% coverage by increasing a to ~15 000 km and i to ~67.5° [6].

At a fixed $a = 15{,}478$km (48-hour orbit), the optimal PDOP occurs at inclinations between 45° and 55°, implying that coverage and geometric accuracy cannot be simultaneously optimized. Spatial analysis further indicates that PDOP ≤ 6 is maintained primarily in the south-polar region, with degraded geometry at mid-to-high northern latitudes.

To improve global uniformity, the North–South ELFO (NS-ELFO) constellation alternates the argument of periapsis between 90° and 270°. This configuration achieves ≥99% global four-satellite coverage with 16 satellites, and global PDOP ≤ 6 when 24 satellites are deployed.
The Circular Lunar Frozen Orbit (CLFO) offers excellent global uniformity, achieving ≥99% global coverage with 16 satellites (four planes), and both global and south-polar PDOP ≤ 6 when 20 or more satellites are used. However, at smaller constellation sizes (e.g., 10 satellites), CLFO underperforms S-ELFO at high southern latitudes, providing only ~50% south-pole coverage at $a = 9{,}000$–10 000 km, compared to >90% for S-ELFO [6].

Beyond geometric metrics, the covariance analysis in the same study revealed that orbit determination accuracy degrades as semi-major axis increases, due to reduced range-rate observability, and that lower inclinations yield better accuracy because of smaller eccentricity and closer apolune distances. The right ascension of the ascending node (RAAN) also strongly influences OD accuracy, especially at low a, due to changing visibility geometry relative to the ground network [7].

Receiver-noise analysis demonstrated that user range error (URE) depends on both orbit determination precision and delay-lock-loop (DLL) thermal noise, the latter scaling with received power; therefore, receiver noise increases at higher a and larger i where apolune signal power decreases.

Finally, transfer trajectory modeling quantified a three-way trade-off among insertion ΔV, trans-lunar injection energy (C3), and transfer duration. Direct transfers achieve short durations (≈4–6 days) but require high ΔV (≈500–1000 m/s), whereas low-energy transfers minimize ΔV (< 300 m/s) at the expense of long durations (> 80 days) and higher C3 magnitudes. Inclinations above 50° yield fewer feasible direct transfers, while larger semi-major axes slightly reduce ΔV requirements due to smaller orbital-energy gaps [6].

3.2 AFS Signal Structure and Standardization

The Augmented Forward Signal is the one-way ranging signal that LANS satellites will broadcast. According to the LunaNet Signal-In-Space Interface specification (LSIS) Volume A – Augmented Forward Signal, the AFS uses a carrier frequency in the S-band (~2491–2492 MHz) [8]. This frequency was selected in line with space frequency coordination recommendations to avoid interference and to differentiate from Earth GNSS L-band signals. The AFS employs a modern spread-spectrum modulation with a dual-channel format: it contains a Data channel that carries navigation messages (ephemeris, clock, etc.) and a Pilot channel (no data modulation) that provides a robust phase reference for signal tracking. This data/pilot dual structure is analogous to signals like GPS L1C or Galileo E1, enabling high-accuracy tracking by allowing longer integration on the pilot and demodulation on the data component. The LNIS/LSIS documents define the modulation scheme (likely DSSS BPSK/QPSK) and the navigation data frame structure for AFS [1]. However, because LunaNet is still evolving, some higher-level definitions – notably the standard navigation message content and the unified lunar time/frame – are still being finalized [23]. As of early 2025, the AFS recommended standard (LSIS-AFS Version 1) has been published, which fixes the signal waveform details, but leaves placeholders for certain message fields (such as how coordinates or time offsets are encoded) [1]. This approach ensures that different organizations can start building compatible transmitters/receivers while the community converges on common reference frames and epoch definitions. The "augmented" aspect of AFS refers to its capability to carry more than just basic PNT data. Studies describe that AFS navigation messages may include additional metadata, for example network service announcements, communications scheduling info, or health/status broadcasts for the LunaNet network [8]. This is unusual compared to terrestrial GNSS signals and is designed to support LunaNet's integrated comm-nav function. For instance, an AFS broadcast could inform a user when to expect a high-bandwidth communication session or relay availability. Balancing these added features with the need for a robust PNT signal has been a focus of signal design. Researchers have factored in lunar-specific challenges such as the spacecraft size/weight/power (SWaP) constraints, which might limit the signal power or clock quality on small lunar satellites [8]. The AFS design is therefore intended to be flexible: it should accommodate varying orbit altitudes and transmitter powers, be resilient to interference/multipath in lunar environments (e.g., coping with surface reflections in crater landscapes), and allow each provider some customization (e.g. regional message augmentation) without breaking interoperability.

Multiple studies and simulations have validated aspects of the AFS and LANS concept. For example, an ESA/NASA/JAXA team analyzed expected user ranging performance given the draft AFS design and found that meter-level User Range Error (URE) can be achieved if the lunar satellites broadcast with sufficient stability and if interoperable time synchronization is in place [19]. There are also ongoing standardization efforts to define a lunar reference frame (akin to WGS84/ITRF on Earth) and a unified lunar time scale to be used by all AFS transmitters [24]. These efforts, under the umbrella of international working groups, are critical for ensuring that a rover can use signals from a NASA satellite and an ESA satellite together seamlessly. In summary, the literature on LANS and AFS depicts a rapidly growing standard.

The Augmented Forward Signal is the key enabling a "lunar GNSS" – it translates the decades of GNSS signal design experience into the lunar context, with adjustments for frequency, content, and interoperability. The next steps will involve flight demonstrations (the first AFS signal transmissions are expected by 2028–29 on test satellites) and iterative refinement of the standard as real-world data is collected.

3.3 Lunar Navigation Receiver and Observation Modeling

Designing lunar navigation receivers and modeling their measurements is another critical research area. Lunar PNT receivers must handle unique challenges: significantly lower signal powers (for Earth-originating GNSS signals), fewer satellites in view during the early phases of LANS, and the absence of Earth-like augmentation (no dense ground monitor network on the Moon). Researchers have extended GNSS measurement models and receiver architectures to the lunar context, covering everything from basic pseudorange and Doppler models to multi-sensor fusion techniques. This section reviews key findings on lunar pseudorange/Doppler modeling, specialized receiver designs, and the integration of IMUs or vision-based sensors to enhance navigation [12].

Pseudorange and Doppler Modeling: The fundamental measurements for satellite navigation – pseudorange (code phase) and Doppler (carrier phase rate) – remain the basis for lunar user positioning. However, the error sources and modeling require adaptation for the Moon's environment. One simplification is that the Moon has no atmosphere, so ionospheric and tropospheric delay errors that affect Earth GNSS are negligible for lunar-based signals. Conversely, other errors become relatively more prominent: satellite clock drift, user clock drift, and multipath from lunar terrain are major contributors to measurement error [13]. High-fidelity lunar simulation studies model pseudorange measurements as the true geometric range plus contributions from satellite clock offset, user clock offset, receiver noise (e.g. thermal noise in code tracking loops), and multipath [13]. For instance, a small navigation satellite in lunar orbit may carry an oven-controlled crystal oscillator or mini-atomic clock (e.g. a miniaturized Rubidium, "MiniRAFS"), whose drift over time adds a bias to the range unless corrected [24]. Similarly, the user's receiver clock introduces its own offset and drift. These clock errors are modeled randomly in simulations, sometimes using clock performance parameters from hardware tests (e.g. JAXA's LNSS studies model satellite clocks as miniRAFS and rover clocks as high-end OCXOs) [15]. Multipath is also a concern: uneven lunar terrain can cause signal reflections that lead to ranging errors. Although the lunar surface is generally radio-quiet (no man-made RFI) and there is no ionosphere, the multipath error can be significant near the surface and must be included in error budgets for precise positioning. The pseudorange-rate (Doppler) measurements are likewise modeled by considering relative user-satellite velocity plus receiver clock drift rate. Doppler shifts on lunar navigation signals can be quite high if the satellite is in low orbit (e.g. a few hundred km altitude orbiter has ~1.6 km/s orbital velocity, causing Doppler on the order of several kHz at S-band). Lunar receivers will employ Phase Lock Loops similar to GNSS receivers, with thermal noise and oscillator phase noise affecting the pseudorange-rate accuracy [15]. One study notes that the dominant sources of phase/Doppler error in lunar receivers are analogous to those on Earth: thermal noise, oscillator phase noise, and dynamic stress error [13]. Overall, the literature shows that with proper modeling of these effects, expected single-measurement accuracy for an AFS pseudorange can be on the order of a few meters (depending on satellite clock quality and multipath environment), and Doppler-based velocity accuracy on the order of cm/s to mm/s for carrier-phase tracking [20].

3.4 Ephemeris parameterization method

There is extensive and growing interest in Earth's Moon, particularly its south pole, which is seen as a potential gateway for future space exploration. To support this, the European Space Agency (ESA) has initiated the "Moonlight" activity, which aims to provide robust communication and navigation services for future lunar infrastructure [31].

A critical component of any navigation system is the design of its broadcast ephemerides—the navigation message that allows users to determine an orbiter's position. This paper investigates two distinct methods for representing these ephemerides for the future Lunar Communication and Navigation Services (LCNS) system [14].

The primary challenge identified by the authors stems from the proposed orbit for the LCNS. Unlike the nearly circular orbits of Earth-based Global Navigation Satellite Systems (GNSS), the lunar orbit is planned to be highly eccentric. This design is necessary to place the aposelene over the lunar south pole, ensuring long-duration coverage for this region of high interest.

However, this high eccentricity introduces significant modeling challenges, particularly when the satellite passes at high velocity through the periselene (the orbit's lowest point), where it is subject to complex gravitational perturbations.

a. Methodology

The authors first simulated a 30-day precise trajectory of a lunar orbiter, subjecting it to a complex force model that included both gravitational and non-gravitational perturbations. This simulated "true" orbit served as the reference for testing the accuracy of two different ephemeris representation methods:

1. Keplerian Elements + Empirical Accelerations: A model consisting of the six classical Keplerian elements supplemented by a set of empirical accelerations (up to nine parameters) to account for perturbations.
2. Chebyshev Polynomials: A model that uses Chebyshev polynomials to fit and represent the satellite's position in X, Y, and Z coordinates.

b. Key Findings and Comparative Analysis

The study provides a trade-off analysis between orbit accuracy, data storage requirements (bit cost), and computational complexity for the end-user.

Comparison Metric	Keplerian Elements + Empirical Accelerations	Chebyshev Polynomials
Best Accuracy (95%)	6.7 cm. Achieved with 6 elements and 9 empirical parameters in a 1-hour window.	3.2 cm. Achieved with a polynomial of degree 10 (11 coefficients) in a 1-hour window.

Comparison Metric	Keplerian Elements + Empirical Accelerations	Chebyshev Polynomials
Bit Storage Cost	Lower: 401 bits total.	Higher: 576 bits (or 595 bits for the 11-coefficient scenario).
User Computation	High Complexity: The end-user must employ a numerical integration algorithm to compute the satellite's position.	Low Complexity: The implementation is simple for the end-user, requiring no numerical integration.
Extrapolation	Possible: The model can be used (with limited accuracy) outside its defined validity window if a message is missed.	Impossible: The model is only valid within its defined time window and cannot be extrapolated.
Partial Message Use	Not applicable.	A key advantage: The user can compute a less accurate position even if only the first few coefficients of the message are received.

c. Conclusions

The authors conclude that both methods are potentially viable for representing the navigation message for the designed LCNS system [11].

The final choice represents a critical system-design trade-off:

- The Keplerian + Empirical Accelerations model is more bit-efficient (401 bits) and robust against missed messages (extrapolation), but it places a significant computational burden on the end-user's receiver.

- The Chebyshev Polynomials model offers higher potential accuracy (3.2 cm) and is computationally simple for the user, but it comes at the cost of higher data storage (576+ bits) and has no extrapolation capability

3.5 Frames to be used in the project

3.5.1 Lunar Inertial Frame (LIF)

The Lunar Inertial Frame (LIF) (also called a Moon-centered inertial frame) is a right-handed, Cartesian frame centered at the Moon's center of mass, with axes fixed in inertial space. This means the LIF does not rotate with the Moon; its orientation remains fixed relative to distant stars or the International Celestial Reference Frame (ICRF). In practice, one can define the LIF axes by aligning them with an Earth-based inertial frame (e.g. J2000 equatorial axes) or the Moon's principal axes at a reference epoch. For example, a common choice is a Moon-centered frame parallel to Earth's mean equator and equinox of J2000 (ICRF); the X-axis points toward the vernal equinox direction (projected from the Moon's center), Z-axis toward the Earth's north celestial pole, and Y-axis completes a right-handed triad. Another choice aligns the inertial axes initially with the Moon's principal inertia axes. In all cases, the LIF is non-rotating with respect to the stars [5].

Applications:

The LIF is used whenever inertial dynamics are required – notably in orbit propagation and high-fidelity trajectory modeling. Spacecraft orbits around the Moon are integrated in the LIF so that Newton's laws apply without fictitious forces. It's also used as an intermediate frame in tracking and navigation: spacecraft ephemerides are often produced in a Moon-centered inertial frame, and then converted to other frames for specific uses. In mission design and deep-space navigation, the LIF provides a stable reference for linking the Moon's motion to the Earth-Sun system. In addition, an inertial frame is needed for orbit determination, where observations are used to estimate a spacecraft's state around the Moon.

Coordinates and Standards:

In LIF, a satellite's state is given by inertial Cartesian coordinates (X,Y,Z and velocities) relative to the Moon's center. There is no particular "shape" associated; however, a standard lunar gravitational parameter is used (GM≈4.9028×10^12 m^3/s^2) for equations of motion. The LIF can be seen as the lunar analogue of Earth's ECI (Earth-centered inertial) frame. Efforts are underway to formalize a "Lunar Celestial Reference System (LCRS)" by adapting IAU/IERS definitions. For now, many simulations simply use Moon-centered J2000 coordinates.

3.5.2 Selenodetic Coordinate System

The selenodetic coordinate system defines positions on and near the Moon's surface in terms of latitude (φ), longitude (λ), and height (H) above a reference sphere or ellipsoid. It is the lunar counterpart of Earth's geodetic system.

By IAU convention, the origin is at the Moon's center of mass, the Z-axis points toward the lunar north pole, and the prime meridian (0° longitude) passes through the mean Earth direction on the Moon's near side. Latitude is measured north (+) or south (–) from the lunar equator, and longitude is measured east-positive (0–360° E). The standard reference radius is 1737.4 km, corresponding to the Moon's mean radius.

This coordinate system is used for mapping, surface navigation, and landing site specification. It allows every surface point to be uniquely described by (ϕ, λ, H) and easily converted to lunar body-fixed Cartesian coordinates for orbital or visibility computations.

Mathematical Definition:

In a Moon-fixed (LBF) Cartesian frame, a point's spherical coordinates are defined as follows. Let (x, y, z) be Cartesian coordinates (origin at lunar center, z along the north pole, $x$toward 0°longitude, y toward 90°E). Then

$$\mathrm{lat}\ (\phi) = \arctan\left(\frac{z}{\sqrt{x^2 + y^2}}\right)$$

$$\mathrm{lon}\ (\lambda) = \mathrm{atan2}(y, x)$$

$$r = \sqrt{x^2 + y^2 + z^2}, H = r - R_{ref} \tag{3.1}$$

If an ellipsoidal reference is used, the latitude conversion requires the standard iterative geodetic solution (usually unnecessary for the Moon). To reconstruct Cartesian coordinates from latitude, longitude, and height:

$$x = (R_{ref} + H)\cos\phi \cos\lambda,$$

$$y = (R_{ref} + H)\cos\phi \sin\lambda,$$

$$z = (R_{ref} + H)\sin\phi \tag{3.2}$$

with $R_{ref} = 1737.4$ km for a spherical lunar reference.

Inter-conversions:

Converting between selenodetic (lat,lon) and other frames often involves two steps: (1) convert lat, lon, height to LBF Cartesian (using the formulas above); (2) rotate or translate to the target frame (e.g., to inertial frame or local ENU). For instance, to get an inertial state of a lunar surface asset, one would take its known lat=ϕ, lon=λ, height=H, compute LBF position (x, y, z), then multiply by the rotation matrix for LBF→LIF at that time to get an inertial position. This is needed if, say, one wants to calculate the Doppler shift of a signal from a lunar base to an orbiter: the base's velocity in inertial space can be obtained from the time-derivative of the coordinate transform (which due to Moon's rotation is roughly 4.6 m/s at equator, since 2πR/27.3 days ≈ 4.6 m/s). In practice, because the rotation is slow, many navigation simulations simply keep everything in the Moon-fixed frame for the user solution, as noted in one study: user positions can be expressed in Cartesian or lat/lon relative to Moon's center, assuming the Moon fixed frame as quasi-inertial for short signal durations.[32]

3.5.3 Radial–Along-Track–Cross-Track Frame (RSW)

The RSW frame is a local orbit frame defined with respect to a spacecraft's instantaneous orbit geometry. It is essentially the same as the RIC (radial-intrack-crosstrack) or RTN (radial-transverse-normal) frame commonly used in orbital mechanics. "RSW" denotes Radial, along-

Track (or along-flight), and Cross-Track (W) axes. This coordinate system is spacecraft-centered and moves with the spacecraft in its orbit. The axes are constructed as follows:

R-axis (Radial):

Points from the central body (Moon) outward through the spacecraft. It is aligned with the spacecraft's position vector r, so

$$R = \hat{r} = \frac{r}{|\, r \,|} \tag{3.3}$$

For a circular orbit, this direction corresponds exactly to the local vertical (pointing away from the Moon's center). For an elliptical orbit, the R-axis still points toward the Moon's center — not always opposite to the instantaneous gravity vector if perturbations exist — but conceptually it represents the radial direction.

S-axis (Along-Track):

Lies in the spacecraft's orbital plane, perpendicular to R, and generally points in the direction of motion. It is essentially the direction of the spacecraft's velocity projected within the orbital plane. Formally, S can be defined using the cross-product to ensure orthogonality:

$$\hat{s} = \frac{w \times r}{|\, w \times r \,|} \tag{3.4}$$

,where $w = r \times v$

In a near-circular orbit, S is nearly aligned with the velocity vector. In an elliptical orbit, particularly near periapsis, the velocity is not exactly perpendicular to the radius, so S deviates slightly from the instantaneous velocity direction but still lies tangent to the trajectory. Some literature refers to this as the transverse or in-track (I) axis.

W-axis (Cross-Track):

Points along the orbital angular momentum vector, effectively normal to the orbital plane. It is defined as

$$W = \frac{r \times v}{|\, r \times v \,|} \tag{3.5}$$

For a prograde orbit around the Moon, W roughly points toward the Moon's north pole. In equatorial orbits, W aligns exactly with the pole, whereas in inclined orbits, it tilts accordingly. Some references denote this axis as N (normal) or C (cross-track); here the symbol W is used for consistency with common orbital mechanics conventions.

By construction, R, S, W are orthogonal unit vectors forming a right-handed triad. R is radially outward, W is along orbit normal (approximately "up" out of plane), and S completes the triad pointing in the general direction of motion (forward along the orbit). If the orbit is exactly circular, S aligns exactly with velocity and W is exactly orthogonal to both position and velocity.

Applications:

The RSW frame is widely used in orbital relative motion analysis and perturbation modeling. In lunar navigation, one application is the parameterization of orbit ephemerides or maneuvers. For example, when fitting an orbit with empirical accelerations, it's common to express small accelerations in RSW directions (as these align with intuitive directions: radial perturbations change altitude, along-track perturbations affect orbital period, cross-track perturbations affect inclination). Indeed, a proposed lunar navigation satellite ephemeris message uses a Keplerian orbit plus 9 empirical acceleration parameters in the R, S, W directions. The reason is that many forces (like drag or SRP or unmodeled gravity) have predictable signatures in these axes (e.g. a primarily along-track acceleration to mimic unmodeled lunisolar perturbations).

Coordinate Relations:

If r and v are the spacecraft's position and velocity vectors in an inertial frame (e.g., LIF), the RSW unit vectors are defined as follows:

$$\hat{\mathrm{u}} = \hat{\mathrm{R}} = \frac{\mathrm{r}}{|\,\mathrm{r}\,|}, \tag{3.6}$$

which points from the Moon's center to the spacecraft.

$$\hat{\mathrm{w}} = \hat{\mathrm{W}} = \frac{\mathrm{r} \times \mathrm{v}}{|\,\mathrm{r} \times \mathrm{v}\,|}, \tag{3.7}$$

which points along the orbital angular momentum vector (normal to the orbital plane).

$$\hat{s} = \hat{S} = \hat{w} \times \hat{u}, \tag{3.8}$$

which yields a unit vector perpendicular to both W and R, completing a right-handed orthogonal system.

For circular orbits, S points in the direction of motion; for non-circular orbits, it follows the instantaneous velocity component orthogonal to R.

These unit vectors can be assembled into a rotation matrix [T] that transforms from the inertial frame to the RSW frame:

$$[T] = \begin{pmatrix} \hat{u}^T \\ \hat{s}^T \\ \hat{w}^T \end{pmatrix},$$

$$[T]\,\Delta r_{\text{inertial}} = \begin{pmatrix} \Delta R \\ \Delta S \\ \Delta W \end{pmatrix} \tag{3.9}$$

so that [T] projects any inertial vector (such as a relative position or acceleration) into the RSW coordinate components $(\Delta R, \Delta S, \Delta W)$. [33]

Examples:

In a ephemeris message for a lunar orbiter, the broadcast could include "empirical accelerations: a_R, a_S, a_W" that the user applies to improve orbit propagation. For instance, if an orbiter has a small bias in the along-track direction due to solar radiation pressure, an a_S term can model it. The user's receiver, when propagating the orbit, will form the RSW frame at each integration step to add these accelerations in the correct direction. Another example: when two satellites in lunar orbit perform a rendezvous, the chaser's guidance might be done in the target's RSW frame – e.g., "move 10 km ahead along-track and 2 km above (cross-track)". The control software will transform those into inertial displacements. Similarly, station-keeping for a frozen lunar orbit might analyze perturbation growth in radial vs cross-track directions to design maneuvers. Because RSW aligns with deviations (radial errors affect altitude, cross-track errors affect plane), it is intuitive for orbit maintenance.

Chapter 4

4 Methodology

4.1 System Overview

This project implements an end-to-end lunar navigation simulation framework, including signal generation, transmission, reception, and navigation solution estimation. The system is composed of two primary components: (1) a Lunar Augmented Navigation Signal simulator that generates Augmented Forward Signal (AFS) broadcasts from multiple lunar satellites, and (2) a software-defined receiver that processes the simulated signals to produce a navigation solution (position, velocity, and timing). The overall flow begins with defining the satellite signal sources and orbits, then generating the AFS signals as they would be transmitted in space, and finally running a receiver algorithm on the simulated radio-frequency data to compute the user's position and time offset. Figure 1 provides a conceptual overview of this process.

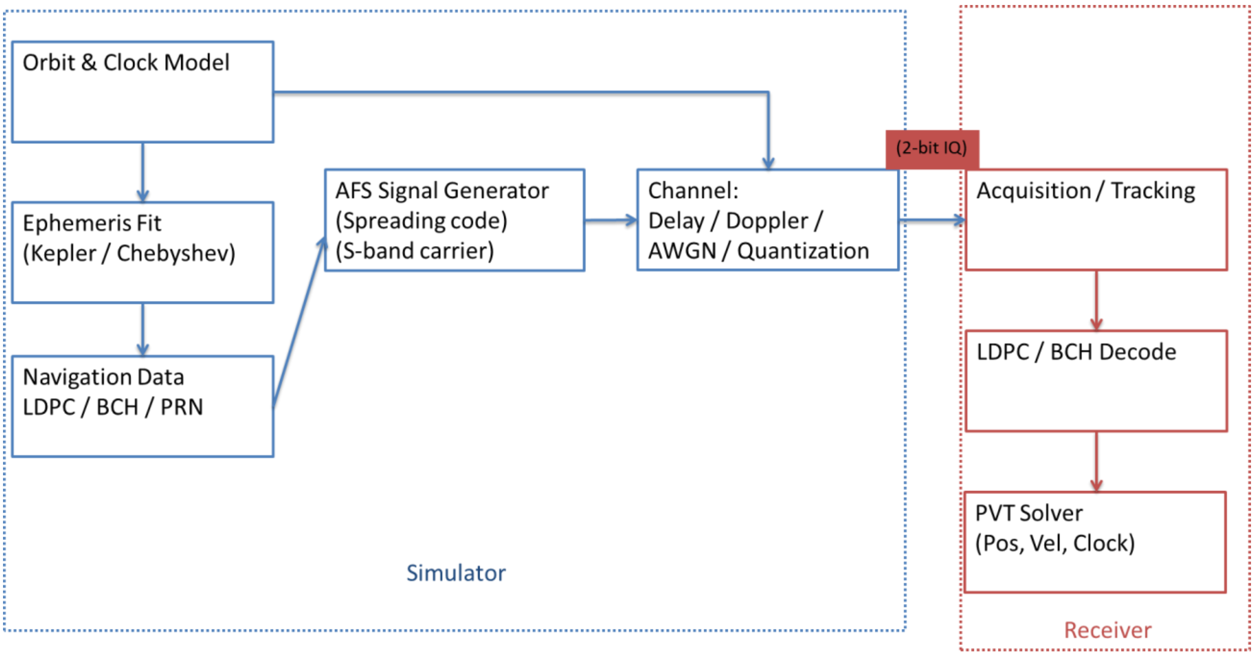

Figure 1 System Architecture

In the simulation, we assume a constellation of lunar navigation satellites broadcasting AFS signals and a single lunar user receiver (e.g., on the lunar surface) capturing those signals. The signal generation module creates digital baseband representations of the AFS waveform for each satellite, incorporating the spreading codes, modulation, and navigation message data as specified by LANS AFS. These signal streams are then combined (summing contributions from all visible satellites) to form the composite RF signal at the receiver input. For simplicity, ideal line-of-sight propagation is assumed – no atmospheric delays and no multipath reflections are modeled. The receiver module then digitizes this composite signal (simulated directly as digital I/Q samples) and performs the standard GNSS processing chain: acquisition of each satellite's signal, code/carrier tracking, demodulation and decoding of the navigation message, and finally position, velocity, and time (PVT) solution computation. Key signal source assumptions include synchronized signal transmission (satellite clocks are modeled with known offsets) and a static or slow-moving user. The system is designed such that the entire flow, from signal generation to navigation solution, can be executed in software for testing and development purposes.

Overall, this methodology enables a standalone AFS-based navigation solution to be developed and validated. In absence of any real AFS transmissions (since LANS is not yet deployed), the simulator provides the necessary test signals. The receiver can thus be developed and tested against a known "ground truth" scenario, where the satellite positions and user location are predefined. This chapter details each part of the methodology: the measurement models and navigation equations used, the design of the AFS signal simulator, the receiver signal processing architecture, and the implementation environment and tools.

4.2 Observation Model and Navigation Equations

a. Pseudorange Measurement Model:

In analogy to Earth GNSS, the fundamental measurement is the pseudorange, which is the apparent range between a satellite and the user as measured by the signal's time of flight. For a given satellite i, the pseudorange ρ_i is modeled as equation(4.1):

$$\rho_i = |r_i - r_u| + c(\Delta t_u - \Delta t_i) + \varepsilon_{\rho,i} \tag{4.1}$$

where r_i is the position of satellite i (at the signal transmission time) in a lunar reference frame, r_u is the user's position (at reception time) in the same frame, c is the speed of light, Δt_u is the receiver clock bias, Δt_i is the satellite's clock bias, and $\varepsilon_{\rho,i}$ represents measurement noise (and other unmodeled errors). This equation indicates that the measured range equals the true geometric range plus the error in receiver clock (which is unknown and common to all satellites) minus the satellite clock error (which is typically known or corrected from the navigation message). In our simulation, atmospheric delays are neglected (no ionosphere or troposphere on the Moon), and relativistic effects (e.g. gravitational or second-order Doppler shifts) are ignored as they are small for our purposes. Additionally, since we adopt a Moon-centered inertial frame, we do not include any rotation-induced Sagnac effect in the range calculation. Under these assumptions, the pseudorange simplifies to the line-of-sight distance plus clock offset terms. After generating the signal, the simulator knows the true ρ_i for each satellite; the receiver will estimate ρ_i by determining the signal's time of transmission and comparing it to the reception time.

b. Doppler (Range-Rate) Measurement Model:

The receiver's tracking loops also provide a Doppler measurement, which corresponds to the rate of change of the range. Doppler is observed as a frequency offset of the carrier caused by relative motion between satellite and user and by any receiver clock frequency error. The relationship between the range rate $\dot{\rho}i$ and the Doppler frequency $f_{D,i}$ is given by:

$$f_{D,i} \approx -\frac{\dot{\rho}_i}{\lambda} = -\frac{1}{\lambda}\frac{d}{dt}|r_i - r_u| \tag{4.2}$$

where λ is the signal wavelength (approximately 0.12 m for the AFS S-band frequency). The negative sign indicates that if the range is increasing (satellite and user receding from each other), the received frequency is lower than the nominal frequency. In practice, the receiver derives $f_{D,i}$ from the carrier tracking loop (e.g., the instantaneous NCO frequency in the PLL). This measured Doppler can be used to compute or aid velocity solutions. For example, given multiple satellites, one can form linear equations in the user velocity components and receiver clock drift. In our implementation we log Doppler measurements for analysis and incorporate them in an extended solution filter (described below), though the primary navigation solution is obtained from pseudoranges. The simplified Doppler model above assumes no significant frequency offsets other than relative motion; any satellite clock drift is typically handled by applying broadcast drift parameters, and receiver clock drift is estimated as part of the navigation state. Equation (4.2) is consistent with the notion that Doppler shift is proportional to range-rate (e.g., $\Delta f = -\Delta\rho/\lambda$).

c.Navigation Solution Equations:

To determine the user's position, we use the set of pseudorange measurements to solve for the unknown receiver coordinates and clock bias. At a given epoch (time of measurement), the system of equations can be written for N visible satellites (with $N \geq 4$ for a 3D fix) by equation(4.3):

$$\left\{ \begin{array}{c} \rho_1 = \sqrt{(x_1 - x_u)^2 + (y_1 - y_u)^2 + (z_1 - z_u)^2} \; + \; c\,\Delta t_u \; - \; c\,\Delta t_1, \\ \rho_2 = \sqrt{(x_2 - x_u)^2 + (y_2 - y_u)^2 + (z_2 - z_u)^2} \; + \; c\,\Delta t_u \; - \; c\,\Delta t_2, \\ \vdots \\ \rho_N = \sqrt{(x_N - x_u)^2 + (y_N - y_u)^2 + (z_N - z_u)^2} \; + \; c\,\Delta t_u \; - \; c\,\Delta t_N, \end{array} \right. \tag{4.3}$$

Where (x_i, y_i, z_i) are the coordinates of satellite i and $(x_u, y_u, z_u, \Delta t_u)$ are the unknown user coordinates and clock bias. These equations are nonlinear in the user's position. The receiver solves them using an iterative least-squares algorithm (Gauss-Newton method): starting from an initial guess (e.g., the previous solution or an estimate), the pseudorange equations are linearized around the guess, forming $H\Delta x \approx \Delta\rho$ where H is the Jacobian (geometry matrix of unit line-of-sight vectors and a clock term) and Δx is the correction to the state. We then solve for Δx in a least-squares sense and update the estimate. This process repeats until convergence. The result is an estimate of (x_u, y_u, z_u) and Δt_u. The receiver clock bias Δt_u effectively absorbs any common mode error in all ranges (including the fact that the receiver's time base might not be perfectly aligned to the system time). By solving for Δt_u, we obtain a timing solution (often reported as a clock offset or as a corrected receive time). In our navigation solution, we output the user's 3D coordinates in the lunar frame and the receiver clock bias. If multiple epochs are processed, a simple numeric differentiation or a Kalman filter can produce the user's velocity and clock drift.

d. Lunar Coordinate and Time System:

Since a standardized lunar reference frame and time system are still being defined, we state the assumptions used in our simulation. We adopt a selenocentric Moon-centered inertial J2000 coordinate system: the origin is at the Moon's center of mass, and the axes are non-rotating with respect to distant stars (for simplicity, we align them to the Moon's principal axes at simulation start). We assume the Moon is a perfect sphere and non-rotating in our scenario, effectively making the Moon-fixed frame an inertial frame. (This avoids needing to account for Moon rotation during signal transit; in reality the Moon does rotate very slowly – one revolution per month – which would cause negligible Sagnac effect over the signal transit time.) User positions are thus expressed in Cartesian coordinates (or latitude/longitude if needed) relative to the Moon's center. For time, we use the GPS time scale as a proxy for a future "Lunar Reference Time". GPS time is a continuous atomic time (synchronized to TAI with a constant offset) with a weekly rollover; using it provides a convenient stable timescale. In practice, our simulator tags signal transmission times using GPS time, and the navigation messages contain time-of-week and week number fields similar to GNSS. This is a stand-in for the eventual LunaNet time standard [2] that is expected to be defined for lunar navigation. All clock biases Δt are thus relative to GPS time in our simulation. These assumptions (non-rotating spherical Moon, GPS time) can be refined once official lunar reference frames and time standards are available.

4.3 UERE and UERRE

Position accuracy in a satellite navigation system depends on both measurement errors and satellite geometry. We define the geometry (design) matrix G as an $n \times 4$ matrix for n satellites (rows) and 4 unknowns (3-D position plus clock bias). In an Earth-centered frame, each row j of G contains the unit line-of-sight direction cosines from the user to satellite j and a 1 for the clock bias term: for example, equation(4.4)

$$G_j = \left[-\frac{x^j - x}{\rho^j},\ -\frac{y^j - y}{\rho^j},\ -\frac{z^j - z}{\rho^j},\ 1 \right] \tag{4.4}$$

where (x, y, z) is the user position, (x^j, y^j, z^j) the jth satellite position, and ρ^j the geometric range. The matrix G is thus the "geometry matrix" that encapsulates satellite-user geometry [25].

Given G, the covariance of the least-squares position solution (ignoring atmospheric biases) is proportional to the inverse matrix by equation(4.5)

$$Q = (G^T G)^{-1} \tag{4.5}$$

The diagonal elements of Q are the variances of the estimated coordinates $(\sigma_x^2, \sigma_y^2, \sigma_z^2, \sigma_t^2)$. The Dilution of Precision factors are defined in terms of these variances. In particular, the Position DOP is calculated by equation(4.6)

$$PDOP = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2} = \sqrt{Q_{11} + Q_{22} + Q_{33}} \tag{4.6}$$

and the Geometric DOP (including time) is calculated by equation(4.7)

$$GDOP = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2 + \sigma_t^2} = \sqrt{Q_{11} + Q_{22} + Q_{33} + Q_{44}} \tag{4.7}$$

Equivalently, $GDOP = \sqrt{PDOP^2 + TDOP^2}$, where $TDOP = \sqrt{\sigma_t^2}$. Low values of PDOP (near 1) indicate a "good" satellite geometry (wide angular spread), while large PDOP values indicate poor geometry and hence large position errors for a given range error. These expressions follow directly from the geometry matrix: one computes $Q = (G^T G)^{-1}$ and then sums its diagonal entries as above.

a. User Equivalent Range Error (UERE)

The User Equivalent Range Error (UERE) represents the combined standard deviation of all pseudorange error sources (as seen by the user). By definition, UERE is the root-sum-square of individual range error components.

The User Equivalent Range Error (UERE) at time t is obtained by combining the space-segment error and the user-segment error as the root-sum-square (RSS) of the Signal-in-Space Error (SISE) and the User Equipment Error (UEE). In other words, the total ranging error experienced by the user is modeled as the quadratic combination of navigation-signal broadcast errors and receiver-induced measurement errors, reflecting both satellite-side and user-side contributions to the overall pseudorange uncertainty by equation(4.8).

$$\mathrm{UERE}(t) = \sqrt{\mathrm{SISE}_{\rho}(t)^2 + \mathrm{UEE}_{\rho}(t)^2} \tag{4.8}$$

b. User Equivalent Range-Rate Error (UERRE)

The User Equivalent Range-Rate Error (UERRE) represents the combined standard deviation of all range-rate (i.e., Doppler-based) error sources perceived by the user. Analogous to the UERE definition in the pseudorange domain, the UERRE at time t is obtained as the root-sum-square (RSS) of the space-segment range-rate error and the user-equipment range-rate error. Specifically, it is formed by quadratically combining the Signal-in-Space range-rate error ($\mathrm{SISE}_{\dot{\rho}}$) and the user-receiver-induced range-rate error ($\mathrm{UEE}_{\dot{\rho}}$), thus capturing both satellite-originated dynamic and timing uncertainties and receiver-side carrier-tracking and propagation-induced distortions. In essence, UERRE provides a consolidated metric for the velocity-projection measurement accuracy as experienced by a lunar PNT user.

$$\mathrm{UERRE}(t) = \sqrt{\mathrm{SISE}_{\dot{\rho}}(t)^2 + \mathrm{UEE}_{\dot{\rho}}(t)^2} \tag{4.9}$$

c. User Position Error Estimation

The overall position error (1σ uncertainty) for the user is obtained by amplifying the range error by the geometry factor. In fact, it can be shown that the RMS position error σ_{pos} is

$$\sigma_{\mathrm{pos}} = \mathrm{PDOP} \times \mathrm{UERE}(t) \tag{4.10}$$

Here σ_{R} is the UERE (range error standard deviation) and PDOP is as defined above. This relation follows from the linear covariance analysis: since each range error of std. dev. σ_{R} contributes equally to each coordinate, the variances in $Q = (G^{\mathrm{T}}G)^{-1}$ scale that noise into position variance [26]. In summary, once PDOP is known from geometry and UERE is estimated from system design, one simply multiplies them to get the 1σ user position error.[15]

And in next section we will discuss how to quantify these error sources.

4.4 Error sources model

Lunar Augmented Navigation Service (LANS) with the Augmented Forward Signal (AFS) is analogous to a lunar GNSS, and it inherits many of the error sources familiar from Earth-based GNSS. Below, we identify the key error sources and discuss their effects on positioning/timing, analytical modeling approaches, and how they can be implemented in simulations. In the lunar context, some terrestrial error sources (like atmospheric delays) are negligible, while others require special consideration (e.g., lunar terrain multipath and time synchronization across agencies).

The pseudorange signal error consists of two primary components: the space-segment error and the user-equipment error. The space-segment contribution is modeled as equation(4.11)

$$\mathrm{SISE}_{\rho}(t) = \sqrt{\sigma_{\rho,\mathrm{prop}}(t)^2 + \sigma_{\rho,\mathrm{mdl}}^2 + \sigma_{\rho,\mathrm{clk}}(t)^2 + \sigma_{\rho,\mathrm{grp}}^2} \tag{4.11}$$

which characterizes the broadcast navigation-signal uncertainty originating from the satellite. This includes the propagated orbital error $\sigma_{\rho,\mathrm{prop}}(t)$, the orbit-model fitting error $\sigma_{\rho,\mathrm{mdl}}$, the satellite clock bias error mapped into the range domain $\sigma_{\rho,\mathrm{clk}}(t)$, and the group-delay uncertainty $\sigma_{\rho,\mathrm{grp}}$. Together, these terms reflect the accuracy limits of the satellite ephemeris and timing information delivered to the user.

Complementarily, the receiver-side contribution to pseudorange error is represented as equation(4.12)

$$\mathrm{UEE}_{\rho}(t) = \sqrt{\sigma_{\rho,\mathrm{rec}}(t)^2 + \sigma_{\rho,\mathrm{path}}(t)^2} \tag{4.12}$$

where $\sigma_{\rho,\mathrm{rec}}(t)$ denotes receiver-internal noise and tracking loop errors, and $\sigma_{\rho,\mathrm{path}}(t)$ accounts for propagation-induced effects such as multipath or surface reflections. This term captures the measurement uncertainty arising from receiver hardware, signal processing, and local environment influences.

The range-rate signal error can be decomposed into contributions from the space segment and the user equipment. The space-segment component is expressed as equation(4.13)

$$\mathrm{SISE}_{\dot{\rho}}(t) = \sqrt{\sigma_{\dot{\rho},\mathrm{prop}}(t)^2 + \sigma_{\dot{\rho},\mathrm{mdl}}^2 + \sigma_{\dot{\rho},\mathrm{clk}}(t)^2} \tag{4.13}$$

which accounts for the broadcast dynamics and timing uncertainty of the navigation signal. Specifically, $\sigma_{\dot{\rho},\mathrm{prop}}(t)$ denotes the propagated orbit-velocity error projected onto the line-of-sight, $\sigma_{\dot{\rho},\mathrm{mdl}}$ captures the residual orbit-fit mismatch when modeling the satellite trajectory, and $\sigma_{\dot{\rho},\mathrm{clk}}(t)$ represents the contribution of satellite/receiver clock frequency instability mapped into the range-rate domain. Together, these terms characterize the fundamental accuracy limit imposed by the satellite's ephemeris prediction and frequency standard performance.

In parallel, the receiver-side contribution to range-rate uncertainty is given by equation(4.14)

$$\mathrm{UEE}_{\dot{\rho}}(t) = \sqrt{\sigma_{\dot{\rho},\mathrm{rec}}(t)^2 + \sigma_{\dot{\rho},\mathrm{path}}(t)^2} \tag{4.14}$$

where $\sigma_{\dot{\rho},\mathrm{rec}}(t)$ represents the receiver carrier-tracking noise and loop filter limitations, and $\sigma_{\dot{\rho},\mathrm{path}}(t)$ corresponds to propagation-induced dynamics such as multipath, scattering, or near-surface effects. This expression captures the local measurement fidelity governed by receiver signal processing, oscillator stability, antenna characteristics, and environmental distortions.

4.4.1 statellite clock error

Lunar navigation satellites carry high-stability oscillators (e.g., RAFS or Rb atomic clocks), but residual timing imperfections inevitably contribute to pseudorange error. Since a 10 ns clock bias produces ≈3 m range error, even small uncorrected clock deviations can degrade LANS positioning and timing performance. The navigation message broadcasts clock correction parameters, yet imperfect estimation and on-board stability limitations cause residual clock error, which contributes to the Signal-in-Space Error (SISE).

Accurate time and frequency references are essential for satellite navigation systems, as any deviation in the onboard oscillator directly propagates into pseudorange and Doppler measurements. To model the evolution of satellite clock states, this work adopts the widely-used stochastic oscillator formulation proposed by Zucca and Tavella (2005). The satellite clock state comprises phase bias $b(t)$, frequency bias $\dot{b}(t)$, and frequency aging $a(t)$, forming the three-dimensional state vector $x_{clk}(t) = [b(t), \dot{b}(t), a(t)]^T$. The dynamics are represented as a random-walk and run-time process by equation(4.15):

$$x_{clk}(t_{k+1}) = \Phi(t_{k+1}, t_k) x_{clk}(t_k) + c w_k, \Phi = \begin{pmatrix} 1 & \tau & \tau^2/2 \\ 0 & 1 & \tau \\ 0 & 0 & 1 \end{pmatrix} \tag{4.15}$$

where $\tau = t_{k+1} - t_k$ and w_k denotes the driving noise associated with oscillator instabilities. This structure parallels the position-velocity-acceleration model, but applied to clock phase, frequency, and drift. The process noise covariance incorporates white frequency modulation (WFM) and random-walk frequency modulation (RWFM) components, reflecting the short- and long-term stochastic behavior of precision oscillators. Notably, these noise sources accumulate over time, resulting in increasing uncertainty in clock predictions during broadcast intervals.

The navigation message provides clock correction parameters a_{f0}, a_{f1}, a_{f2}, corresponding to the instantaneous values of $b(t_0), \dot{b}(t_0)$, and $a(t_0)$ at the reference epoch. Between updates, the onboard clock covariance is propagated using the state transition matrix and process noise, analogous to standard Kalman filtering techniques [27]:

$$E(w_k) = \mathbf{0}_{3\times1}, \qquad E(w_k w_k^T) = S_k(t_{k+1}, t_k) = \begin{pmatrix} \sigma_{\mathrm{WFM}}^2 T + \sigma_{\mathrm{RWFM}}^2 \frac{\tau^3}{3} & \sigma_{\mathrm{RWFM}}^2 \frac{\tau^2}{2} & 0 \\ \sigma_{\mathrm{RWFM}}^2 \frac{\tau^2}{2} & \sigma_{\mathrm{RWFM}}^2 \tau & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$P_{clk}(t_{k+1}) = \Phi P_{clk}(t_k)\Phi^T + S_k \tag{4.16}$$

The resulting variance in phase bias directly contributes to pseudorange error $\sigma_{\rho,clk}(t) = \sqrt{P_{11}(t)}$, while the frequency bias variance and short-term oscillator stability (typically quantified via Allan deviation) determine the Doppler or range-rate uncertainty. Consequently, oscillator characteristics represent a fundamental performance driver in lunar PNT systems, especially under extended broadcast durations and long signal-propagation delays. While relativistic effects also influence clock states, they are assumed to be corrected within the navigation data and are not explicitly modeled here.

In velocity estimation based on short observation periods, the oscillator's short-term frequency stability becomes a dominant factor. This stability reflects the ability of the local oscillator to maintain a consistent output frequency over a finite averaging duration. Importantly, such stability is a property of the oscillator itself and does not depend on the nominal carrier frequency. It is commonly characterized by metrics such as the Allan deviation or the Hadamard deviation, denoted $\sigma_y(T)$ and $\sigma_H(T)$, respectively. For a measurement duration T_m, the contribution of oscillator instability to the line-of-sight velocity error can therefore be modeled as equation(4.17)

$$\sigma_{\dot{\rho},clk}(t) = \sqrt{P_{22}(t) + c^2\, \sigma_y^2(T_m)} \tag{4.17}$$

which combines the dynamic clock-state uncertainty $P_{22}(t)$ with the short-term stochastic frequency fluctuation characterized over the interval T_m.[27]

4.4.2 Ephemeris / Orbit Prediction Errors

Accurate satellite ephemerides are essential for precise positioning and velocity determination. In practical navigation systems, the broadcast ephemeris is generated by fitting a finite-order dynamic model or polynomial representation to a high-fidelity reference orbit. This approximation introduces inevitable modeling errors due to imperfect dynamical knowledge, limited parameter resolution, and finite update intervals of the navigation message.

Consequently, the broadcast ephemeris differs from the true satellite trajectory, leading to residual orbit errors that propagate into user range and range-rate measurements.

Let $r_{true}(t)$ denote the true satellite position and $r_{fit}(t)$ the broadcast ephemeris estimate obtained by polynomial or Keplerian fitting. The residual ephemeris error in three-dimensional space is defined as equation(4.18)

$$\Delta r(t) = r_{true}(t) - r_{fit}(t) \tag{4.18}$$

A navigation receiver observes only the projection of this spatial error onto the satellite–receiver line-of-sight (LOS) direction $\hat{u}(t)$. The resulting pseudorange error is therefore given by equation(4.19)

$$e_{\rho,eph}(t) = \hat{u}^T(t)\, \Delta r(t) \tag{4.19}$$

where $\hat{u}(t) = \frac{r_{true}(t) - r_{usr}}{\| r_{true}(t) - r_{usr} \|}$ denotes the LOS unit vector from the receiver to the satellite.

Similarly, taking the time derivative of the ephemeris residual yields the induced pseudorange-rate (Doppler) error,

$$e_{\dot{\rho},eph}(t) = \hat{u}^T(t) \left(v_{true}(t) - v_{fit}(t) \right) \tag{4.20}$$

where $v_{true}(t)$ and $v_{fit}(t)$ are the true and fitted satellite velocities, respectively. These terms jointly quantify the degradation in user position and velocity estimates caused by imperfect broadcast orbit modeling.

In practice, the statistical behavior of ephemeris-fit residuals depends on orbit dynamics, fitting interval, and the chosen representation (e.g., Keplerian parameters, Chebyshev polynomials, or numerical propagation). The resulting errors are commonly characterized by their root-mean-square (RMS) value within each ephemeris segment and subsequently mapped into the overall Signal-in-Space Error (SISE) budget for both range and range-rate domains.

4.4.3 Receiver Noise Analysis

The user range error (URE) introduced by receiver noise arises not only from the satellites' orbit determination (OD) uncertainty [7] but also from the thermal noise inherent in the receiver's Delay Lock Loop (DLL). The DLL thermal noise determines the random error in code tracking and is a function of the received signal power. Following the formulation, the DLL noise variance for satellite k can be expressed as

$$(\sigma_{DLL}^2)_t^{(k)} = \begin{cases} \dfrac{B_{DLL}}{2(C/N_0)_t^{(k)} d}\left(1+\dfrac{2}{T(C/N_0)_t^{(k)}(2-d)}\right), & d \geq \dfrac{\pi}{T_c B_{fe}} \\ \dfrac{B_{DLL}}{2(C/N_0)_t^{(k)}}\left(\dfrac{1}{T_c B_{fe}}\right)\left(1+\dfrac{1}{T(C/N_0)_t^{(k)}}\right), & d \leq \dfrac{1}{T_c B_{fe}} \\ \dfrac{B_{DLL}}{2(C/N_0)_t^{(k)}}\left(\dfrac{1}{T_c B_{fe}}+\dfrac{B_{fe}T_c}{\pi-1}\left(d-\dfrac{1}{T_c B_{fe}}\right)^2\right)\left(1+\dfrac{2}{T(C/N_0)_t^{(k)}(2-d)}\right), & \dfrac{1}{T_c B_{fe}} < d < \dfrac{\pi}{T_c B_{fe}} \end{cases}$$

Here, B_{DLL} denotes the DLL loop bandwidth, T_c is the chip duration, B_{fe} is the double-sided front-end bandwidth, T represents the coherent integration time, and d is the early-late correlator spacing. The carrier-to-noise density ratio $(C/N_0)_t^{(k)}$ is expressed in linear scale (not dB-Hz).

To compute $(C/N_0)_t^{(k)}$ for each satellite, a standard link-budget model is employed by equation(4.21):

$$\left(\frac{C}{N_0}\right)_t^{(k)} = P_{tx} + G_{tx} + G_{rx} - L_{tx}^{(k)} - L_{fs}^{(k)} - 10\log_{10}(k_0 T_{sys}) \qquad (4.21)$$

where P_{tx} is the transmitted power, G_{tx} and G_{rx} are the antenna gains at the transmitter and receiver, respectively, $L_{tx}^{(k)}$ is the transmitter cable loss, $L_{fs}^{(k)}$ is the free-space path loss, $k_0 =$

1.38×10^{-23} J/K is Boltzmann's constant, and T_{sys} denotes the receiver system temperature. The free-space path loss is further given by equation(4.22)

$$L_{fs}^{(k)} = 20 \log_{10}\left(\frac{4\pi d^{(k)} f}{c}\right) \tag{4.22}$$

where $d^{(k)}$ is the satellite–receiver distance, f the carrier frequency, and c the speed of light. The satellite transmit antenna pattern follows Melman et al. (2022) and is scaled according to the apparent lunar-horizon angle at apolune as equation(4.23)

$$G_{tx}(\theta) = G_{ref}(\theta/\alpha), \alpha = \frac{\tan^{-1}\left(\frac{r_{moon}}{r_{ref} + h_e}\right)}{\tan^{-1}\left(\frac{r_{moon}}{r_{moon} + h_e}\right)} \tag{4.23}$$

where $r_{moon} = 1737.4\,\text{km}$is the lunar radius, $r_{ref} = 9750\,\text{km}$the reference distance, $h_e$the satellite altitude, and $G_{ref}(\cdot)$ the reference antenna gain pattern.

Finally, the computed $(C/N_0)_t^{(k)}$ values are substituted into equation(4.21) to estimate the DLL thermal noise variance, which is then converted into pseudorange noise as equation(4.24)

$$\sigma_\rho^{(k)} = c\, T_c \sigma_{DLL}^{(k)} \tag{4.24}$$

In addition to pseudorange tracking noise, the receiver also contributes to pseudorange-rate (Doppler) noise through the Phase-Lock Loop (PLL) or Frequency-Lock Loop (FLL), which determine the stability of carrier tracking. The corresponding receiver-induced range-rate error is modeled as equation(4.25)

$$\sigma_{\dot{\rho},rec}^{(k)}(t) = c\, \sigma_{\dot{\phi}}^{(k)}(t) \tag{4.25}$$

where $\sigma_{\dot{\phi}}^{(k)}(t)$ denotes the PLL/FLL phase-rate standard deviation. For a PLL-based tracker, this can be approximated by equation(4.26)

$$\sigma_{\dot{\rho},rec}^{(k)}(t) = c \sqrt{\frac{B_{PLL}}{2(C/N_0)_t^{(k)} T_m}} \tag{4.26}$$

with B_{PLL} denoting the PLL loop bandwidth and T_m the Doppler measurement interval. This formulation allows consistent propagation of receiver thermal noise into both the pseudorange and pseudorange-rate error domains.[28]

4.4.4 Multipath

When navigation signals travel in proximity to a planetary surface—whether on Earth or the Moon—they may not reach the receiver solely through the direct line-of-sight path. Instead, additional components can arise from reflections off the surface or nearby structures, resulting in signals arriving through longer indirect paths. For time-based observables, such as TOA or RTD measurements, these reflected components can disturb the Delay-Lock Loop (DLL), thereby introducing bias and fluctuations in the ranging estimates.

To account for this effect in lunar navigation scenarios, this work adopts the open-sky multipath model originally proposed by Brenner for Earth-based GNSS environments. The model characterizes multipath-induced ranging errors as a zero-mean Gaussian process, with the error variance governed by the satellite elevation angle. Specifically, the standard deviation of the multipath error is expressed as [28]

$$\sigma_{MPTH} = a + be^{c\phi} \tag{4.27}$$

where σ_{MPTH} denotes the multipath error standard deviation, ϕ is the satellite elevation angle, and a, b, and c are empirical parameters chosen to represent different signal conditions and operational environments.

4.5 Simulator Architecture

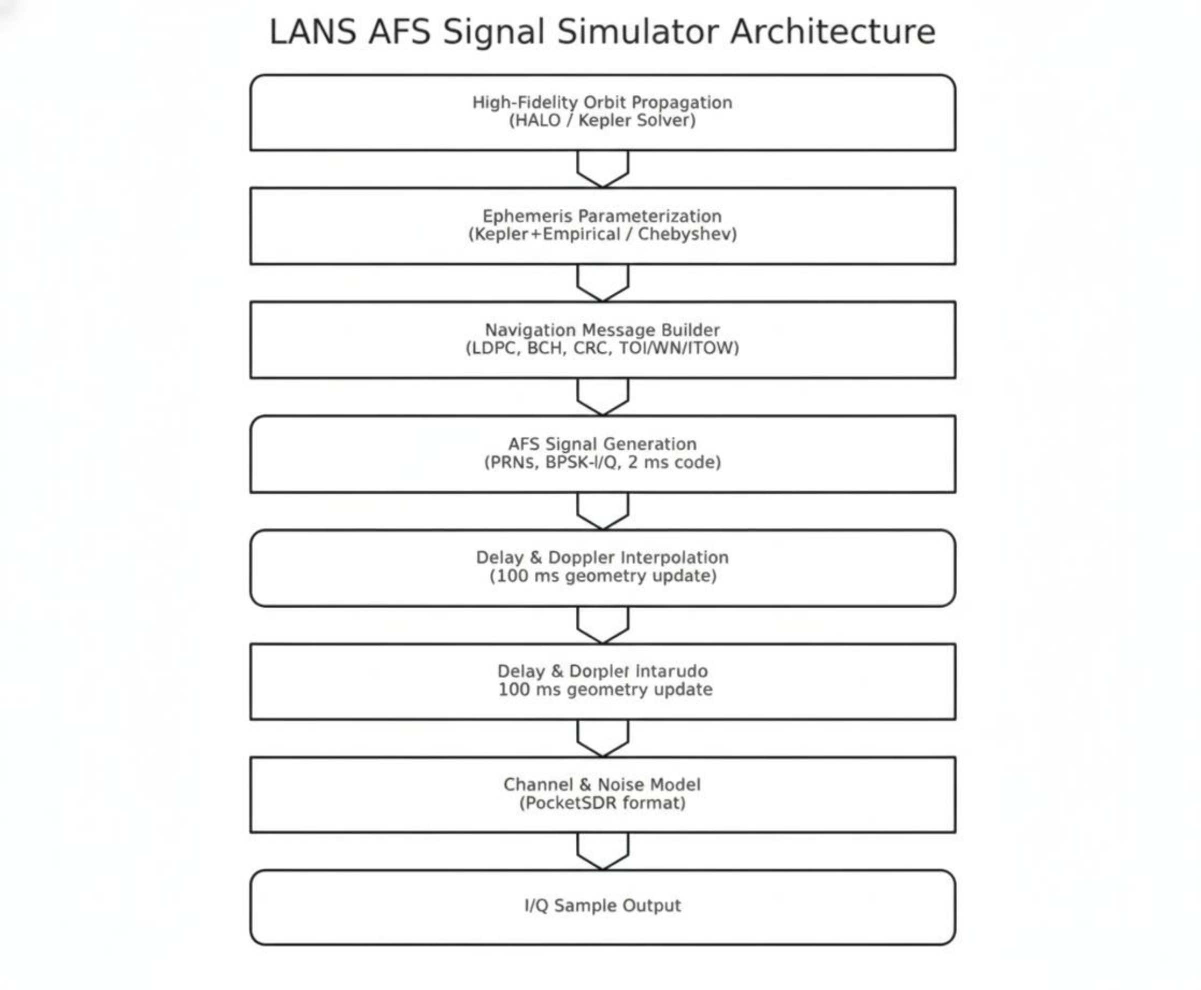

Figure 2 AFS simulator architecture

To generate realistic LANS AFS signals [1], we built a software-based signal simulator by modifying and extending open-source GNSS simulation tools. This simulator is based on the architecture of GPS-SDR-SIM (an open-source GPS L1 C/A signal generator), reconfigured to produce the LANS Augmented Forward Signal [9]. It is capable of generating a digital baseband waveform containing the composite signals from multiple lunar navigation satellites. The output is a stream of interleaved I/Q samples that can be either saved to a file or transmitted via an SDR hardware. Our simulator implementation follows the AFS Signal-in-Space specification (LSIS-AFS) in terms of signal structure (modulations, spreading codes, data frame format), with certain simplifications made for undefined elements (as discussed below).

4.5.1 Orbit and Satellite Motion Modeling

We model the orbits of the lunar navigation satellites using Keplerian orbital elements as initial state of the satellite, allowing various orbit configurations to be simulated. For this research, we selected an Elliptical Lunar Frozen Orbit (ELFO) constellation, as such orbits have been proposed to provide robust coverage of the lunar poles. For example, our baseline scenario uses eight satellites distributed in two orbital planes (four satellites per plane) in ELFO orbits. This high-eccentricity frozen orbit ensures that each satellite spends a long duration with line-of-sight to the south pole when at high altitude, while the periselene is chosen to remain above the lunar surface (periselene altitude ≈ 880 km in our case).

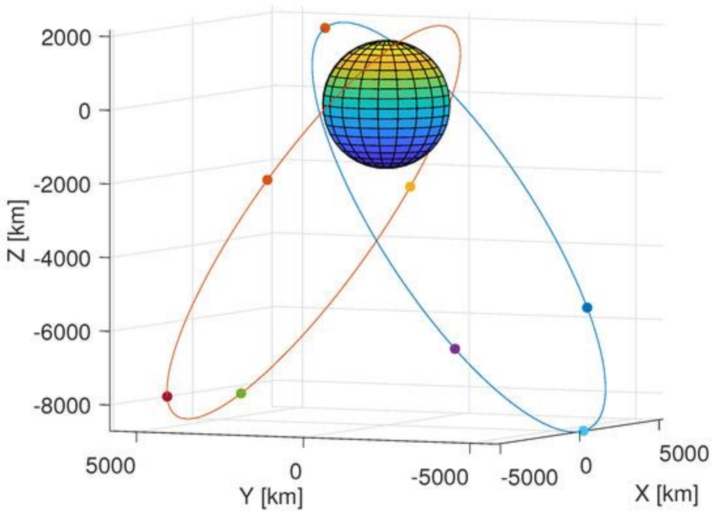

Figure 3 a two-plane s-ELFO constellation

The simulator has the capacity to retrieve each satellite's orbit with two method, one using the two-body Keplerian model (neglecting perturbations for now) and another directly from a csv file. Given an initial state (position and velocity or orbital elements) for each satellite, it computes the satellite's position at each simulation time step by solving Kepler equation. For another method, we can use external tools to propagate a more precise trajectory. In our case, we chose HALO [4] as the orbit propagation tool to generate the orbit within the propagation time.

We assume satellites broadcast their signals continuously and have stable onboard clocks (we can optionally assign small drifts or offsets to simulate clock errors). The user's position is also specified. In our primary scenario, the user receiver is placed at the Shackleton Crater (approximately 89.6°S, 0°E) on the surface – a location of interest due to persistent shadow and potential ice deposits. Using this scenario, the simulator can determine which satellites are above the user's horizon (line-of-sight). We enforce a mask angle (e.g., 5° elevation) to decide visibility; satellites below that angle are ignored. This way, the simulator knows at any given time which satellites' signals should be present.

4.5.2 Ephemeris parameterization

We parameterize broadcast ephemerides for a highly-eccentric Lunar ELFO by evaluating two complementary representations: (i) six Keplerian elements augmented with empirical accelerations in the RSW frame, and (ii) Chebyshev polynomial coefficients of the Cartesian position. Both are derived from a high-fidelity reference trajectory that models lunar/terrestrial gravity (including spherical harmonics), third-body effects, tides, SRP, and small non-gravitational forces, integrated at 1-min steps; windows of 1–4 h are considered for message validity.

A. Keplerian + empirical-acceleration (RSW) model

1. State and force reduction. For each validity window, estimate the classical Keplerian set $(a, e, i, \Omega, \omega, M_0)$ and a 9-parameter empirical acceleration in the radial (R), along-track (S), and cross-track (W) axes:

$$\begin{aligned} R &= R_0 + R_s \sin u + R_c \cos u \\ S &= S_0 + S_s \sin u + S_c \cos u \\ W &= W_0 + W_s \sin u + W_c \cos u, \end{aligned} \tag{4.28}$$

with u the argument of latitude referenced to the Sun. The empirical terms absorb once-per-revolution signatures and secular mismodeling (e.g., SRP, residual gravity terms). Parameters are obtained by least-squares fit to the precise orbit while the user dynamics reduce to a point-mass lunar gravity plus the empirical accelerations.

2. User propagation. At the receiver, recover position/velocity from the Keplerian elements and numerically integrate

$$\ddot{X} = -\mu X/r^3 + a_{emp}(u) \tag{4.29}$$

over the window with a fixed step (e.g., collocation/Runge–Kutta). The RSW unit triad is formed from $X, \dot{X}$ each step to map a_{emp} to LIF.

3. Windowing and accuracy. Sub-meter 95-percentile 3D residuals require the full 6+9 model; with a 1 h window, 95-percentile errors fall to centimeter-level, while longer windows degrade near periselene due to high velocity and geometry changes.

B. Chebyshev Cartesian position model

1. Coefficient estimation. Within each window, fit independent Chebyshev series to $X(t), Y(t), Z(t)$ mapped to $x \in [-1,1]$:

$$X, Y, Z = \sum_{i=0}^{n} a_i\, T_i(x) \tag{4.30}$$

Coefficient degree/window length are tuned for accuracy vs. payload size.

2. Windowing and accuracy. Degree-10–11 (11 coefficients) over 1 h provides ≤ 0.11 m (95-pct) 3D errors; shorter windows or higher degree mitigate periselene degradation. Variable windows tied to velocity (short near periselene, long near aposelene) further stabilise errors.

4.5.3 Signal Modulation

For each visible satellite, the simulator generates the AFS signal that it would transmit. The LANS AFS is a dual-channel, spread-spectrum signal in S-band. Unlike traditional GNSS L-band signals, AFS is centered at ~2492.028 MHz in the S-band. We implement both the Data channel and Pilot channel components, which are in-phase (I) and quadrature (Q) respectively. Each satellite's signal is formed as two channels (figure 3):

$$I_{AFS}(t) = \sqrt{\frac{P_I}{2}} C_{AFS-I}(t) D_{AFS-I}(t)$$

$$Q_{AFS}(t) = \sqrt{\frac{P_Q}{2}} C_{AFS-Q}(t) \quad (4.31)$$

where D_{AFS-I} is the I channel data symbol sequence, P_I and P_Q are the received power levels for I and Q channels, each assumed to be -163 dBW. Figure 4 indicates the process of AFS signal modulation.

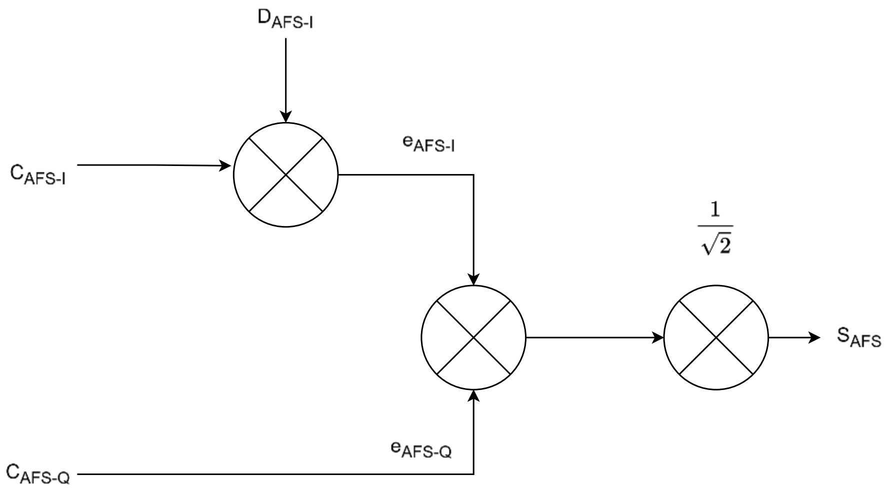

Figure 4 AFS signal modulation diagram

- Data channel (I-component): BPSK-modulated with a pseudorandom noise (PRN) spreading code of length 2046 chips. The PRN code is a Gold code (derived from a 2047 maximum length sequence, with one chip per period removed). The chip rate is 1.023 Mcps (million chips per second), identical to GPS L1 C/A, which yields a code period of 2 ms for the 2046-chip sequence. This short code repeats every 2 ms. Navigation data bits are modulated onto this channel at 500 symbols per second. To achieve robust messaging, the data is encoded with forward error correction: the AFS standard uses a rate-1/2 LDPC (Low-Density Parity Check) code along with interleaving. In our simulator, we implemented the specified LDPC encoder to generate parity bits, and we interleave the codeword bits as per the standard. The LDPC code adds redundancy, which allows the receiver to correct many bit errors in challenging signal conditions.

- Pilot channel (Q-component): BPSK-modulated with a longer PRN code of length 10,230 chips (which is actually the same sequence as the GPS L1C pilot primary code). The pilot chip rate is 5.115 Mcps, so it also has a 2 ms period (10,230 chips at 5.115 MHz). Importantly, the Pilot carries no data symbols – it is a pure unmodulated spreading code, potentially overlaid with a slower secondary code sequence for long integration. In AFS, the pilot has a tiered code structure (primary 2 ms code with secondary/tertiary codes on top). We include the primary code and a secondary code as defined in the draft standard for simulation purposes. The pilot's role is to assist acquisition and tracking under weak signal conditions, since the receiver can integrate the pilot over many milliseconds (even seconds) without worrying about data bit transitions.

The two channels are quadrature-multiplexed, meaning the Data PRN modulates the cosine (I) carrier and the Pilot PRN modulates the sine (Q) carrier. The composite AFS signal from one satellite can be expressed as a passband signal $s(t) = I(t)\cos(2\pi f_c t) + Q(t)\sin(2\pi f_c t)$, where f_c is the carrier frequency (~2.492 GHz), $I(t)$ is the baseband data-channel signal (taking values ±1 as per the spread data bits), and $Q(t)$ is the baseband pilot signal (±1 as per the spreading code). In simulation we actually generate it in digital baseband (complex I/Q) at a chosen sampling rate, typically 12 Msps or higher to capture the 5.115 MHz bandwidth of the pilot channel. [1].

4.5.4 Navigation Message Content

Each satellite's Data channel carries a continuous navigation message, which we structure according to the Augmented Forward Signal frame definition. Although the Clock and Ephemeris Data (CED) message format was not fully defined in the initial LSIS-AFS release [1], we constructed a plausible navigation message with the necessary fields for our simulation. We assume a frame period and subframe structure as described in the standard: for example, a main navigation frame composed of multiple subframes, starting with a preamble. In our implementation, each navigation frame begins with a 68-symbol Synchronization Pattern (SP) that is unencoded (no LDPC). This is essentially a known bit sequence that the receiver looks for to achieve frame sync. Following the SP, we include:

- Subframe 1: Contains time information such as an interval count or epoch marker. In the AFS spec, this is the Time of Interval (TOI) or similar, which helps the receiver reconstruct the transmission time. We encode this subframe with a BCH(52,9) code as recommended [1]. In our case, TOI represent a time interval number within the week, e.g., 0–99 (since frame intervals might be predefined). We ensure the encoded bits of Subframe 1 allow the receiver to identify the time of week or an index when decoded.

- Subframe 2: Contains the Clock and Ephemeris Data (CED) for that satellite – analogous to the ephemeris in GPS. We pack the satellite's orbital parameters (at least a partial set sufficient to compute its position) and clock bias/drift parameters into this subframe. This subframe is encoded with the long LDPC code (rate 1/2, length matching the frame). We generated a parity-check matrix for the LDPC based on the specifications and use it to encode the bits.

- Subframes 3 and 4: The AFS standard mentions these for future use (perhaps for augmentation info or additional messages) but currently undefined. In our simulation, we do not transmit any information in subframes 3 and 4 (they could be filled with dummy bits or omitted). The receiver accordingly ignores these in processing.

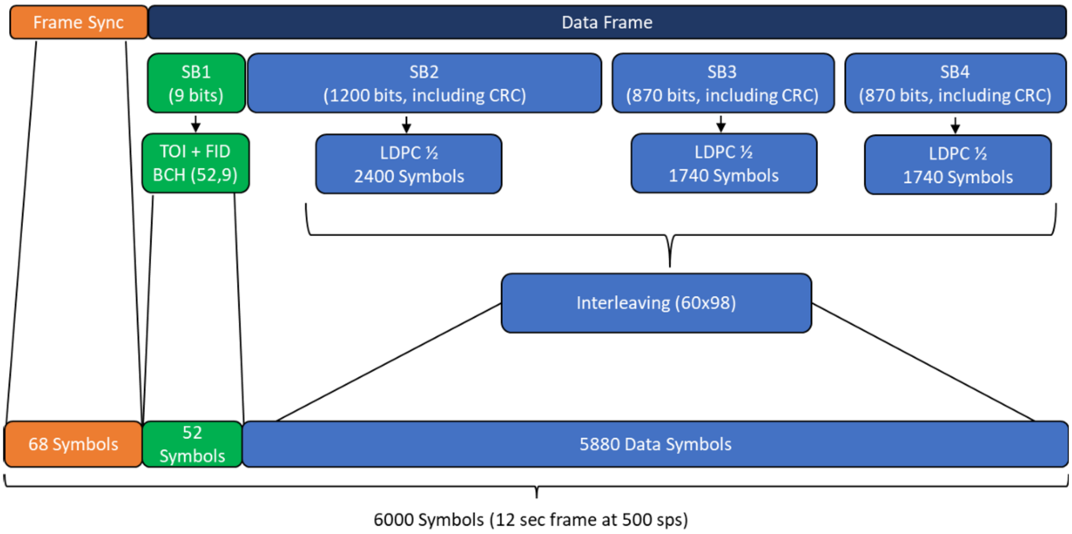

Figure 5 Navigation Message Generic Structure [1]

The frame is repeated continuously, and we assign appropriate cyclic redundancy checks (CRCs) to subframes as needed. For instance, after decoding Subframe 2 (CED), the receiver computes a CRC to verify integrity [1]; our simulator populates the CRC field so that if the receiver bits match, the CRC passes. If a decoding error occurs (e.g., LDPC fails or CRC fails), the receiver will discard that frame and try the next. By structuring the navigation data this way, we ensure the receiver can derive each satellite's clock correction and position once it has decoded a frame.

4.5.5 Simulator Signal Processing

In a digital baseband signal generator, each in-phase and quadrature (I/Q) channel must be sampled at a rate exceeding the bandwidth of the spreading code to preserve signal fidelity. However, continuously calculating the satellite position and corresponding propagation delay at every sampling instant would be computationally prohibitive. To address this challenge, the system evaluates the exact propagation delay only at discrete time steps, while intermediate values are estimated through linear interpolation [10].

The propagation delay $\tau(t)$ at any reception time t is determined from the pseudorange $\rho(t)$ as equation(4.32)

$$\tau(t) = \frac{\rho(t)}{c} \tag{4.32}$$

where c is the speed of light. The pseudorange is defined as equation(4.33)

$$\rho(t) = r(t) - c\,dT \tag{4.33}$$

with $r(t)$ denoting the geometric distance between the satellite and receiver, and dT the satellite clock bias.

Instead of recalculating $\rho(t)$ for every sample, its value at each sampling point t_i is updated by linear prediction:

$$\rho(t_{i+1}) = \rho(t_i) + \Delta\rho\,\delta t \tag{4.34}$$

where δt is the sampling interval and $\Delta\rho$ represents the mean pseudorange rate across the interpolation window ΔT. This rate is derived from equation(4.35)

$$\Delta\rho = \frac{\rho(t_i + \Delta T) - \rho(t_i)}{\Delta T} \tag{4.35}$$

The Doppler frequency shift associated with this motion is then obtained by equation(4.36)

$$\Delta f = -\frac{\Delta\rho}{\lambda} \tag{4.36}$$

where λ denotes the carrier wavelength. In practice, an interpolation interval of $\Delta T = 100$ ms achieves an effective compromise between computational efficiency and accuracy, adequately representing the slow nonlinear variations in delay due to relative motion of the satellite and receiver.

Once the signal transmission time is known, the generator determines the corresponding code phase and data symbol position. The absolute frame time is computed as equation(4.37)

$$t_f = WN \cdot SECWEEK + ITOW \cdot BI_d + TOI \cdot F_d + \Delta t \tag{4.37}$$

where WN is the week count from the LunaNet Reference Time (LRT) epoch, $SECWEEK = 604{,}800$ s is the number of seconds per week, $ITOW$ is the number of 1200-second block intervals (BI_d) elapsed since the start of the week, and TOI denotes the frame index within the current block. Each frame duration F_d corresponds to 12 seconds, or 6000 data symbols.

The parameters WN, ITOW, and TOI are embedded in the Time of Transmission (ToT) message within the navigation data and can be extracted once code tracking is established. The fractional term Δt refines the time within the current frame by combining the symbol-level offset (symbol index multiplied by the 2 ms code period) and the fractional code phase derived from correlation.

The output of the simulator is a stream of raw I/Q samples representing the RF signal at a certain intermediate frequency or at baseband. The simulator outputs 16-bit signed samples by default. This format is common for SDR devices and recordings (e.g., 16-bit IQ). However, our target receiver (PocketSDR) expects 2-bit quantized samples to maximize processing efficiency. Therefore, we include a quantization step: after generating the high-resolution signal, we can downsample or directly quantize to 2-bit. We also add channel noise to simulate a realistic signal. The noise is generated as Gaussian white noise added to each sample, with a specified C/N_0. We typically add noise such that the C/N_0 of the strongest signal is around, say, 45 dB-Hz, but we can adjust this to test the receiver's sensitivity. To generate Gaussian noise efficiently, we use a Central Limit Theorem approach. The noise is scaled to achieve the desired SNR and added to I and Q. We then perform 2-bit quantization: essentially we determine a threshold (close to the standard deviation of the noise to achieve highest C/N_0 accorrding to figure 6 where N=4) and map the combined signal + noise samples to 2-bit values $-1, +1$ on each of I and Q, with 2 bits total representing 4 levels). This mimics a real GNSS front-end ADC, and results in about 0.5 dB SNR loss due to quantization, which is negligible in navigation performance. By doing this, our output file can be fed directly into PocketSDR, which is optimized for 2-bit inputs.

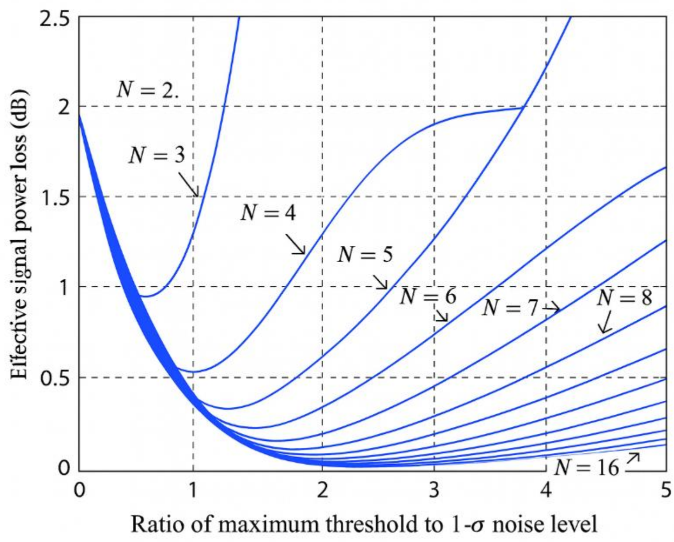

Figure 6 C/N_0 degradation using an N-level quantization [29]

4.6 Receiver Architecture

The receiver is a software-defined GNSS receiver customized for the Lunar AFS signal. We base our implementation on the open-source PocketSDR-AFS framework [9], which provides a lightweight yet efficient AFS receiver pipeline. PocketSDR is written in C and optimized for running on embedded platforms. This version has extended PocketSDR (creating a version we refer to as PocketSDR-AFS) to handle the LANS AFS signal structure. This involved adding the AFS spreading code generators, adjusting acquisition/tracking parameters for the AFS modulation, and implementing a new navigation message decoder for the AFS frames.

4.6.1 Signal Acquisition

The first stage in the receiver processing is acquisition, where the goal is to detect each satellite's signal and obtain rough estimates of its Doppler frequency and code phase. Our acquisition algorithm performs a two-dimensional search over code delay and Doppler frequency for each satellite PRN code. We generated the PRN codes for AFS (both Data and Pilot) using the known generator polynomials – PocketSDR's code generation routines were easily adapted to produce the 2046-chip and 10,230-chip codes. For acquisition, we take segments of the incoming 2-bit IQ data (e.g., 10 ms or 20 ms) and correlate them with a locally generated replica of the satellite's code, mixing the signal with a range of trial Doppler frequencies. Because AFS has a pilot channel without data modulation, we can exploit it for acquisition: we perform a longer coherent integration on the pilot (since it has no bit flips) to boost sensitivity. Typically, we use a coherent integration of 2 ms (one code period) times a noncoherent integration of several accumulations to detect the correlation peak. A detected peak above the threshold yields the satellite ID, an approximate code phase (time delay), and Doppler shift. In practice, we implemented a slightly simplified search: we assume the receiver has a rough idea of user time (within a few seconds) so that the Doppler search space can be limited based on expected satellite-user relative speeds (which are at most a few hundred m/s for lunar orbits, translating to Doppler offsets in the order of ±1 kHz). Acquisition outputs are then passed as initial conditions to the tracking loops. The DFT-based acquisition process, illustrated in Figure 7, follows the parallel code phase search algorithm. The procedure can be summarized as follows:

1. A 2 ms segment of the received digital signal s_n is first downconverted to near-baseband, producing w_n.
2. This sequence is Fourier-transformed and complex-conjugated to yield $X^*(\theta)$.
3. The locally generated PRN code $c_{n+\delta}$ is multiplied by a complex exponential $\exp\{j2\pi f_i(n+\delta)T_s\}$, introducing a Doppler frequency shift.
4. The resulting sequence is then Fourier-transformed, giving $Y(\theta)$.
5. The two spectra are multiplied to obtain the frequency-domain correlation:

$$R_k(\theta) = X^*(\theta) \cdot Y(\theta) \tag{4.38}$$

6. $R_k(\theta)$ is converted back to the time domain via an inverse FFT, yielding $r_k(\delta)$.
7. The maximum correlation amplitude, $\max(|r_k(\delta)|^2)$, is compared to a detection threshold λ.
8. If the threshold is exceeded, acquisition is declared successful—the code phase is identified by δ, and the Doppler frequency estimate is f_i.

9. Otherwise, a new Doppler bin f_i is selected, and the algorithm repeats from step 1.

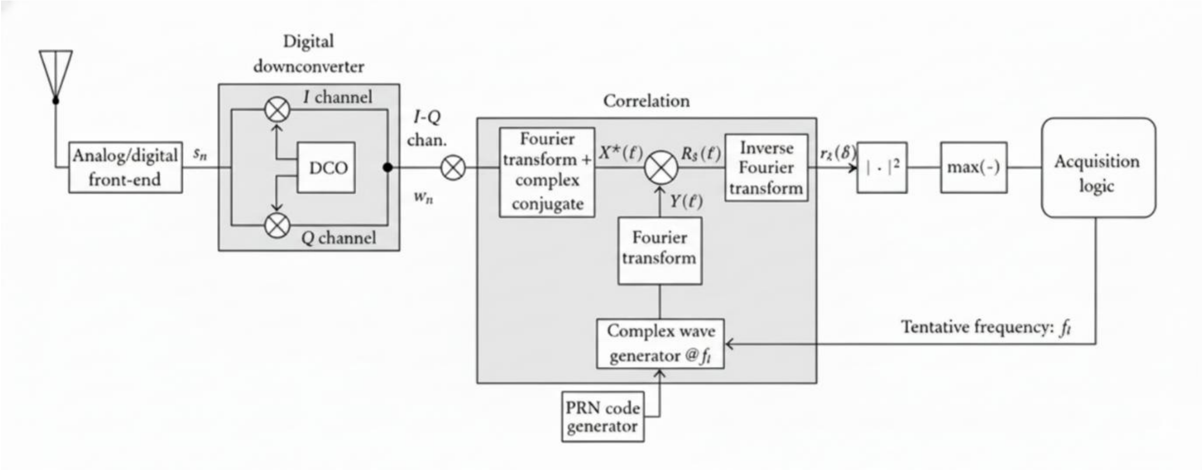

Figure 7 DFT-based acquisition algorithm

4.6.2 Signal Tracking

Once a satellite's presence is confirmed, the receiver enters the tracking stage for that signal. We allocate a tracking channel for the satellite, which performs two main functions: code tracking and carrier tracking. We employ a Delay-Locked Loop (DLL) to track the code phase. This is typically an early-prompt-late correlator that adjusts the code phase NCO to keep the prompt correlation at peak. We use a narrow correlator spacing to improve precision. Simultaneously, a Phase-Locked Loop (PLL) tracks the carrier frequency and phase. Because AFS has a pilot channel, our design can use a pilot-assisted PLL: the pilot, being data-free, allows an extended coherent integration (e.g., 10 ms or more) which makes carrier tracking more robust at low C/N_0. In the current implementation, we actually instantiate two tracking channels per satellite (one for Data, one for Pilot) in PocketSDR, each with its own correlators, but the pilot's carrier estimate can aid the data channel's loop. The tracking loops output fine estimates of pseudorange (from the DLL's code phase) and Doppler (from the PLL's NCO frequency) continuously. These are the raw observables needed for the navigation solution. Under nominal conditions, tracking on both channels remains stable. If the data channel's SNR drops, the receiver can still maintain lock via the pilot. Figure 8 illustrates the diagram of tracking loop algorithm involving DLL and PLL.

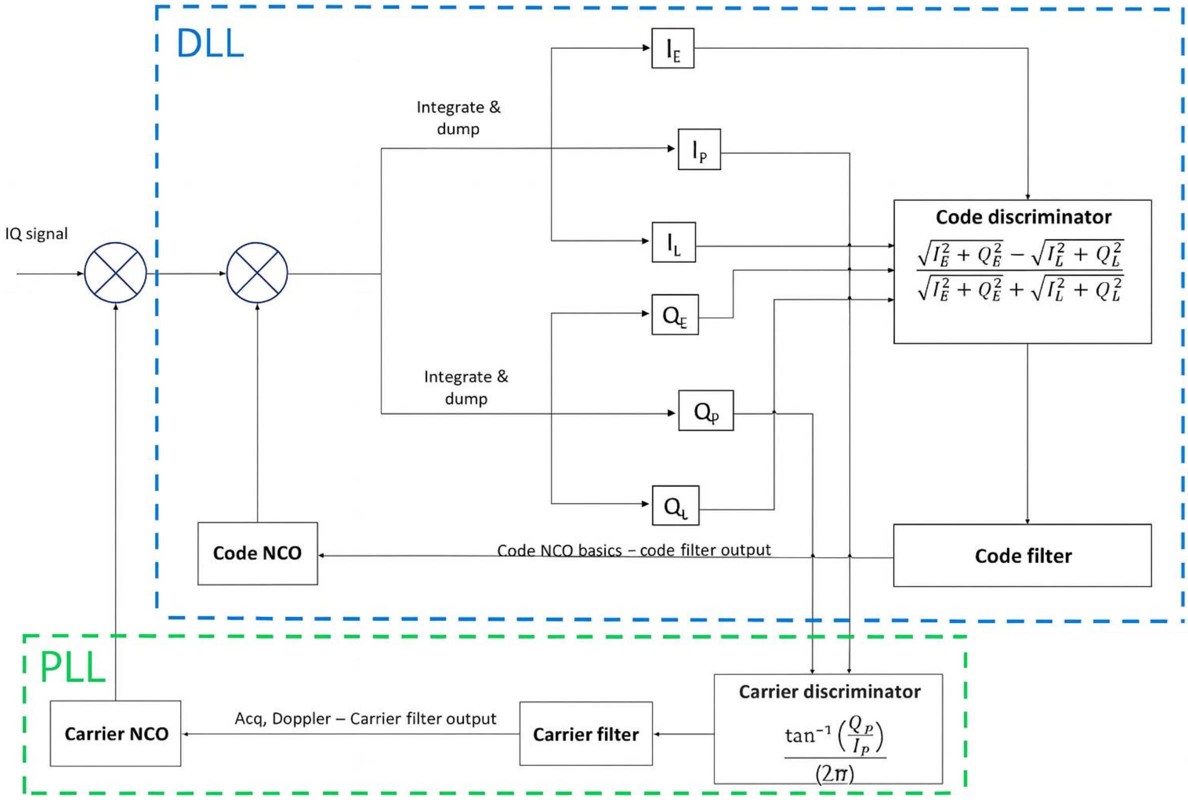

Figure 8 Tracking loop algorithm

4.6.3 Navigation Data Decoding

While tracking is ongoing, the receiver accumulates the demodulated symbols from the Data channel to reconstruct the navigation message. Each symbol is exactly one 2 ms code period, and symbol timing is aligned by design.[1] This means once we are tracking code lock, we also know the symbol boundaries (no separate bit synchronization is needed, unlike legacy GNSS where one must find bit edges). The receiver buffers the incoming symbols in a frame buffer. Using the known 68-symbol Synchronization Pattern (SP) at the start of each frame, we perform frame synchronization. Concretely, our software continuously checks if the last 68 decoded symbols match the predefined SP sequence. When a match is found, it predicts where the next frame's SP should appear (one frame length apart) and verifies it. Upon confirmation, we declare a locked frame. Now the receiver knows the exact framing of the navigation data.

Next, the receiver decodes the subframes in the navigation message. Subframe 1 (which contains the TOI or time information) is protected by a BCH code.[1] We implemented a small BCH(52,9) decoder to decode this, but as an optimization, we can determine the TOI by correlation against all possible codewords. Either way, from Subframe 1 we obtain an integer time tag (e.g., which 2-minute interval of the week this frame corresponds to). Subframe 2 contains the Clock and Ephemeris Data (CED) and is LDPC-encoded. We integrated an LDPC decoder in the receiver, reusing much of PocketSDR's existing CNAV-2 (GPS L1C) decoder architecture. The main adaptation was using the parity-check matrix specific to LANS AFS (we used the same matrix as our simulator's LDPC encoder) [1]. The decoder is iterative, using e.g. a belief propagation algorithm to correct errors in the received bits. We also handle punctured bits: in the AFS LDPC design, some bits may be omitted to improve efficiency, and the receiver treats those as erasures with equal probability of 0/1 (zero log-likelihood). Our implementation inserts zeros for those LLRs so that the decoder knows it must infer them. The LDPC decoding yields the full CED message (satellite orbit and clock data). We then perform a CRC check on the decoded CED to ensure it's correct. If the CRC passes, we accept the data; if it fails (or if the LDPC decoder did not converge), the receiver will discard that frame and wait for the next, continuing to track in the meantime.

Once a frame is successfully decoded, we immediately use the information: from Subframe 1/TOI we know the transmission time of the start of the frame; from Subframe 2 we get the satellite's orbital parameters and clock bias. Combining these with the measured code phase provides a precise pseudorange. Specifically, the receiver computes the signal's Time of Transmission (TOT) by taking the frame's reference time (e.g., the time of the first symbol of the frame, derived from TOI and week number) and adding the offset for the particular symbol and code phase where the measurement was made. PocketSDR uses the formula given by the AFS spec to do this time reconstruction. The result is a transmit timestamp for each satellite's signal. Subtracting that from the reception time (which the receiver has in its clock) and multiplying by c yields the raw pseudorange. We then apply the satellite clock correction (from the CED message) to this pseudorange, and are left with a corrected pseudorange that ideally equals the geometric range plus receiver clock bias. Similarly, the Doppler frequency from tracking is mapped to a pseudorange rate measurement (after accounting for the carrier wavelength and any known clock drift).

4.6.4 Navigation Solution Computation

With multiple pseudoranges now available (one per tracked satellite with decoded data), the receiver proceeds to compute the user's navigation solution. We developed a lunar positioning solver integrated into the PocketSDR code. A custom routine takes the satellite positions and pseudorange measurements and performs an iterative least-squares solution for $(x_u, y_u, z_u, \Delta t_u)$ in the Moon-centered frame. This solver assumes the same spherical Moon model that we used in the simulation. The algorithm is the standard one outlined earlier in Section 4.2: form the geometry matrix from satellite-unit vectors, iteratively solve for the receiver's position and clock bias. We typically start the iteration with an initial guess at the center of the Moon or a rough known location of the landing site if available. In simulation, we often know the true answer (for evaluation), but the solver does not use the truth except as an initial guess if we choose. After a few iterations, it converges to the correct position. The solution gives the user's Cartesian coordinates in the lunar MI frame, which can be converted to latitude/longitude if needed. The clock bias is given as well (in seconds); multiplying by c gives the user clock error in meters or the bias needed to align the user clock to GPS time.

Clock Bias Estimation: The receiver clock bias Δt_u is estimated as part of the navigation solution. We do not have an external time reference (since this is a standalone receiver), so we treat Δt_u as an unknown in the equations and solve for it. Essentially, the solution provides the difference between the receiver's internal time and the GPS time that the satellites are using. In our simulation, the receiver's time is initialized arbitrarily (it starts counting from 0 when the run begins), so Δt_u initially is large (on the order of the true GPS time of reception). The solution will yield a value, which means the receiver's clock is that many seconds off from GPS time. We apply this bias in reporting the navigation time. In a real scenario, the user could then steer its clock or just use the bias in time computations. Because we do not simulate any other timing aids, the clock bias is determined solely by the satellite signals.

4.6.5 Weak Signal Handling

The LANS AFS signal is designed with robust features to handle weak signals (e.g., at lunar surface the signals might be low power). Our receiver leverages some of these features. Firstly, the presence of the pilot channel allows for long coherent integrations in acquisition and tracking; if the signals are weak (low C/N_0), we can integrate over many code periods on the pilot to pull the signal above noise. In our software, we can dynamically extend the coherent integration time in tracking based on C/N_0 estimates – effectively narrowing the loop bandwidth when signals are weak, which improves sensitivity. Secondly, the powerful LDPC error correction in the navigation message means we can tolerate a high bit error rate and still recover the data correctly. Our tests indicate that even at C/N_0 around 20–25 dB-Hz, the LDPC decoder can often decode the message. This is important for urban or obstructed scenarios on the Moon (e.g., a rover in a ravine) where signals could be attenuated. We also implemented pilot-only tracking where if the data channel lock is lost, we continue to track the pilot (which has no data bit reversals) to maintain carrier lock, and later re-acquire the data when it improves. Currently, our position solution uses only the data channel's pseudorange (since that's where the precise code phase is measured and aligned with message decode, but in principle we could combine pilot and data measurements to get an even stronger composite measurement. Using the pilot to aid the data tracking is a planned enhancement.

Chapter 5

5 Simulation and Results

5.1 Simulation Setup

In our simulation, we used the modified open-source LANS-AFS-SIM signal generator and the PocketSDR-AFS receiver framework [9]. The target user was placed near the lunar south pole (e.g. Shackleton Crater, ~88°S latitude) [24]. We assumed a small constellation of 3–5 orbiters. The test case used two polar ELFO planes (8 satellites total) with semi-major axes ~6000–6500 km and high eccentricity (e≈0.6) for long dwell times over the pole, which is shown in table 1.

PRN	SMA (km)	ECC	INC (°)	LAN (°)	AOP (°)	MA (°)	Time
PRN-01	6540.000	0.600000	56.300	0.000	90.000	0.000	1984 May 30 16:44:48.000
PRN-02	6540.000	0.600000	56.300	0.000	90.000	90.000	1984 May 30 16:44:48.000
PRN-03	6540.000	0.600000	56.300	0.000	90.000	180.000	1984 May 30 16:44:48.000
PRN-04	6540.000	0.600000	56.300	0.000	90.000	-90.000	1984 May 30 16:44:48.000
PRN-05	6540.000	0.600000	56.300	180.000	90.000	45.000	1984 May 30 16:44:48.000
PRN-06	6540.000	0.600000	56.300	180.000	90.000	135.000	1984 May 30 16:44:48.000

PRN	SMA (km)	ECC	INC (°)	LAN (°)	AOP (°)	MA (°)	Time
PRN-07	6540.000	0.600000	56.300	180.000	90.000	-135.000	1984 May 30 16:44:48.000
PRN-08	6540.000	0.600000	56.300	180.000	90.000	-45.000	1984 May 30 16:44:48.000

Table 1 Each satellite's initial state in Kepler orbital elements

Key simulation parameters included a 12 MHz sampling rate (16-bit IQ) and additive white Gaussian noise to mimic thermal background. Data sets ranged from seconds to minutes long (e.g. 90 s files) for offline tests. We used a full-sky elevation mask (≈0°) to count line-of-sight satellites. The user was assumed static on the surface with a high-quality MiniRAFS clock. Here lists representative parameter values:

- Sampling rate: 12 MHz (complex IQ).
- Observation interval: 90 s (typical offline test duration).
- Elevation mask: 5° (line-of-sight to Moon limb).
- Noise model: AWGN on I/Q (2-bit quantization).
- User location: Shackleton Crater (88°S, 11.4°E) on the lunar South Pole.

5.2 Constellation analysis

Figure 9 illustrates the long-term orbital evolution of an Elliptical Lunar Frozen Orbit (ELFO) over a 10-year numerical propagation period. The semi-major axis remains tightly bounded around ~6540 km with small quasi-periodic oscillations, demonstrating long-term stability under the coupled effects of lunar gravity harmonics and third-body perturbations. The eccentricity exhibits a periodic variation between approximately 0.60–0.70, while the inclination oscillates smoothly around ~55° with multi-year secular modulation driven primarily by solar and Earth gravitational influences. The right ascension of the ascending node shows continuous precession, consistent with the intrinsic dynamical properties of high-inclination lunar orbits. The argument of periapsis reveals a repeating long-period libration pattern, and the mean anomaly increases nearly linearly with expected periodic resets. Overall, the trajectory maintains bounded orbital elements without secular divergence, confirming the frozen-orbit characteristics of ELFO and its suitability for sustained lunar navigation service deployment over decadal timescales.

Tables 2 and 3 summarise the coverage performance of the simulated eight-satellite ELFO constellation over the lunar south polar region. As shown in Table 2, each individual satellite provides approximately 73–76% visibility, corresponding to 9.7–10.0 hours of average daily visibility and revisit intervals of roughly 3.2–3.5 hours. This reflects the highly elliptical nature of the orbits, where satellites spend extended periods near apolune with favourable geometry over the south pole.

At the constellation level (Table 3), coverage performance is significantly strengthened by orbital phasing: single- and dual-satellite visibility are guaranteed 100% of the time, ensuring continuous availability of at least two navigation links. Higher-order diversity is also robust, with quadruple coverage achieved 99.384% of the time and six-fold coverage maintained for 70.34% of the observation period. This indicates that the constellation provides persistent multi-satellite visibility suitable for robust geometric dilution of precision (GDOP) and resilient lunar PNT services in the south polar region.

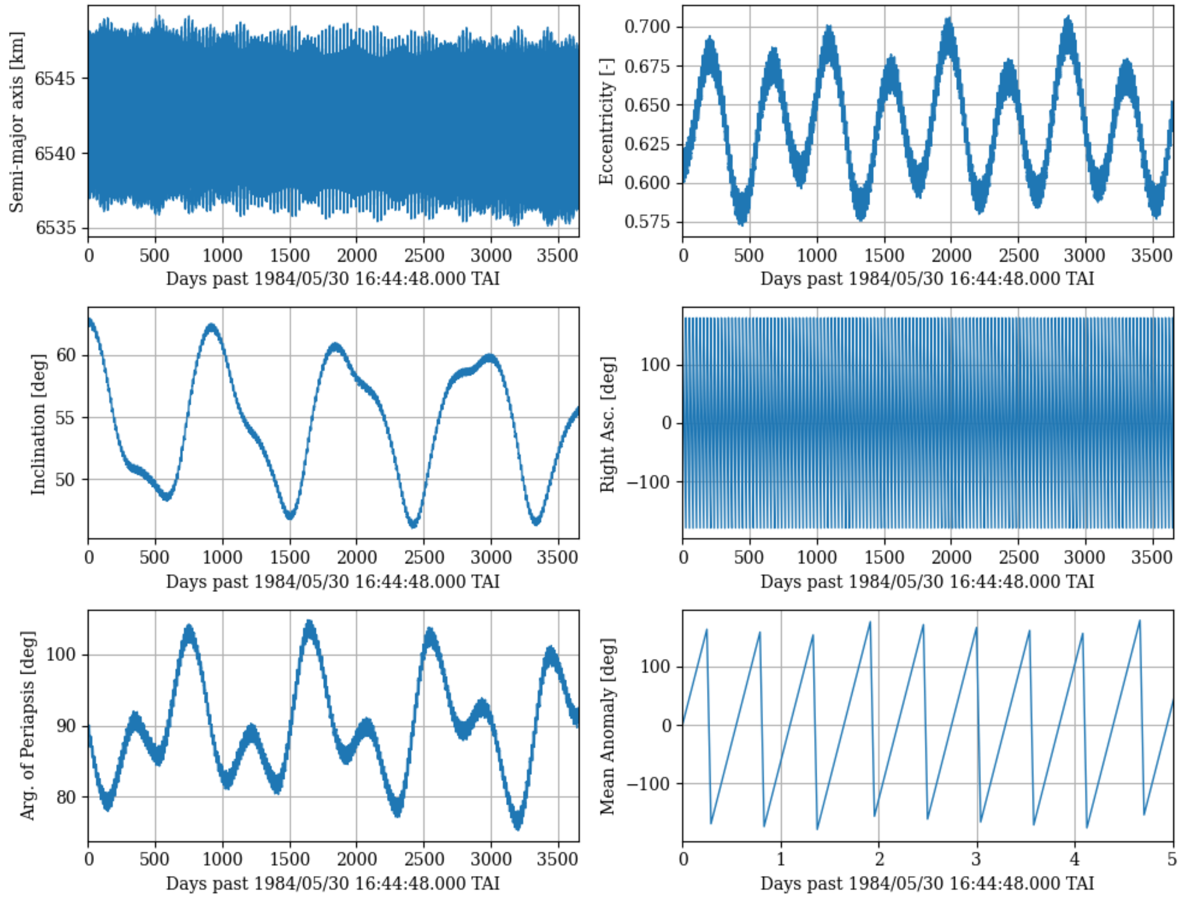

Figure 9 ELFO long-term evolution(10 years)

PRN	Coverage (%)	Avg. Visible Duration (hr)	Avg. Interval (hr)
01	73.68	9.698	3.467
02	73.67	9.697	3.468
03	73.64	9.707	3.478
04	73.67	9.696	3.468
05	75.76	9.964	3.191
06	75.73	9.975	3.200

PRN	Coverage (%)	Avg. Visible Duration (hr)	Avg. Interval (hr)
07	75.71	9.972	3.202
08	75.73	9.967	3.197

Table 2 each satellite’s coverage over lunar south pole(2 years)

Coverage Level	Visible Satellites	Availability (%)
Single coverage	≥ 1	100.000
Dual coverage	≥ 2	100.000
Quadruple coverage	≥ 4	99.384
Six-fold coverage	≥ 6	70.340

Table 3 coverage analysis(2 years)

Figure 10 illustrates the Position Dilution of Precision (PDOP) performance for the proposed eight-satellite ELFO constellation over a two-year simulation at the lunar South Pole. Overall, the system sustains consistently low PDOP values, typically ranging between 1.5 and 3.0, indicating strong geometric diversity and reliable positioning capability in the vicinity of the Shackleton region. Two recurring PDOP peaks are observed near approximately 150 days and 480 days, corresponding to periodic constellation geometry alignments that briefly reduce satellite visibility diversity as orbital nodes precess. Despite these temporary degradations, the system maintains PDOP values below 5 for the vast majority of the mission duration, with only short intervals showing higher spikes. Importantly, the constellation consistently satisfies the minimum three-satellite visibility requirement, ensuring uninterrupted position observability. These results demonstrate that the optimized 8-satellite architecture provides robust and stable geometric coverage at the lunar South Pole with near-continuous low-PDOP performance suitable for high-accuracy PNT services under the LunaNet framework.

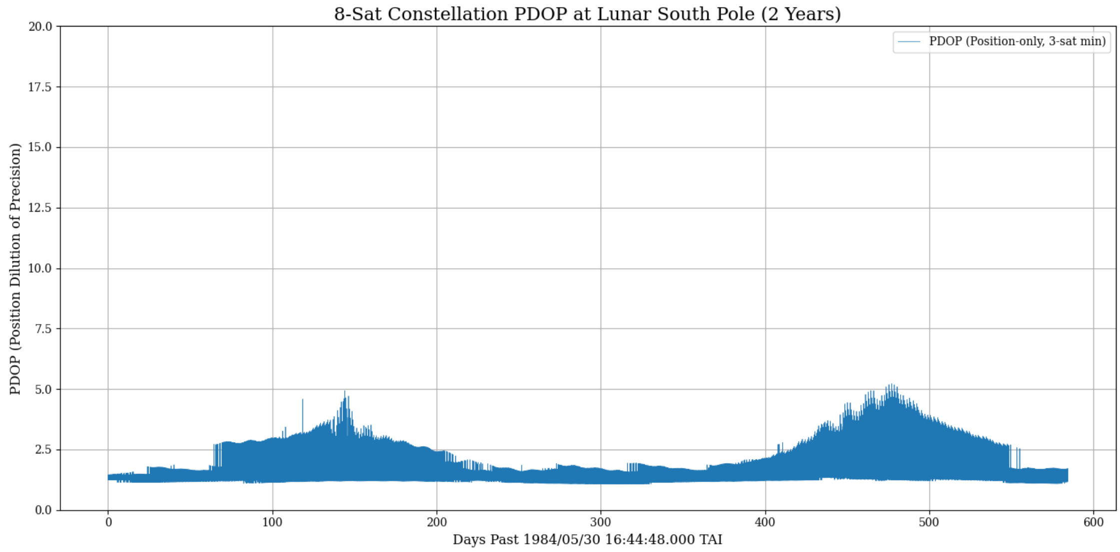

Figure 10 8-Sat constellation PDOP

5.3 Ephemeris comparison

To assess the feasibility of compact broadcast ephemerides for lunar ELFO satellites, we evaluated Chebyshev polynomial fits of varying orders (5, 7, 9, and 11) against high-fidelity HALO truth trajectories. A fixed 1-hour fitting window was used for all satellites, consistent with the AFS broadcast update cycle. Figure 11 compares the resulting position errors for each PRN and the RMS across the constellation.

Lower-order Chebyshev fits (orders 5 and 7) exhibit noticeable approximation bias, with peak errors reaching approximately 100 m and 3 m respectively. These oscillatory errors align with the highly elliptical halo orbit dynamics, where non-Keplerian perturbations from the Earth–Moon multi-body environment are not fully captured by low-degree polynomials. Increasing to order 9 yields a substantial improvement, reducing the RMS error to sub-decimeter level and limiting peak deviations to below 0.10 m throughout the 1-hour window. The best performance is observed with an 11th-order fit, where RMS error falls below 1 cm, satisfying stringent deep-space PNT accuracy targets and demonstrating excellent stability across satellites.

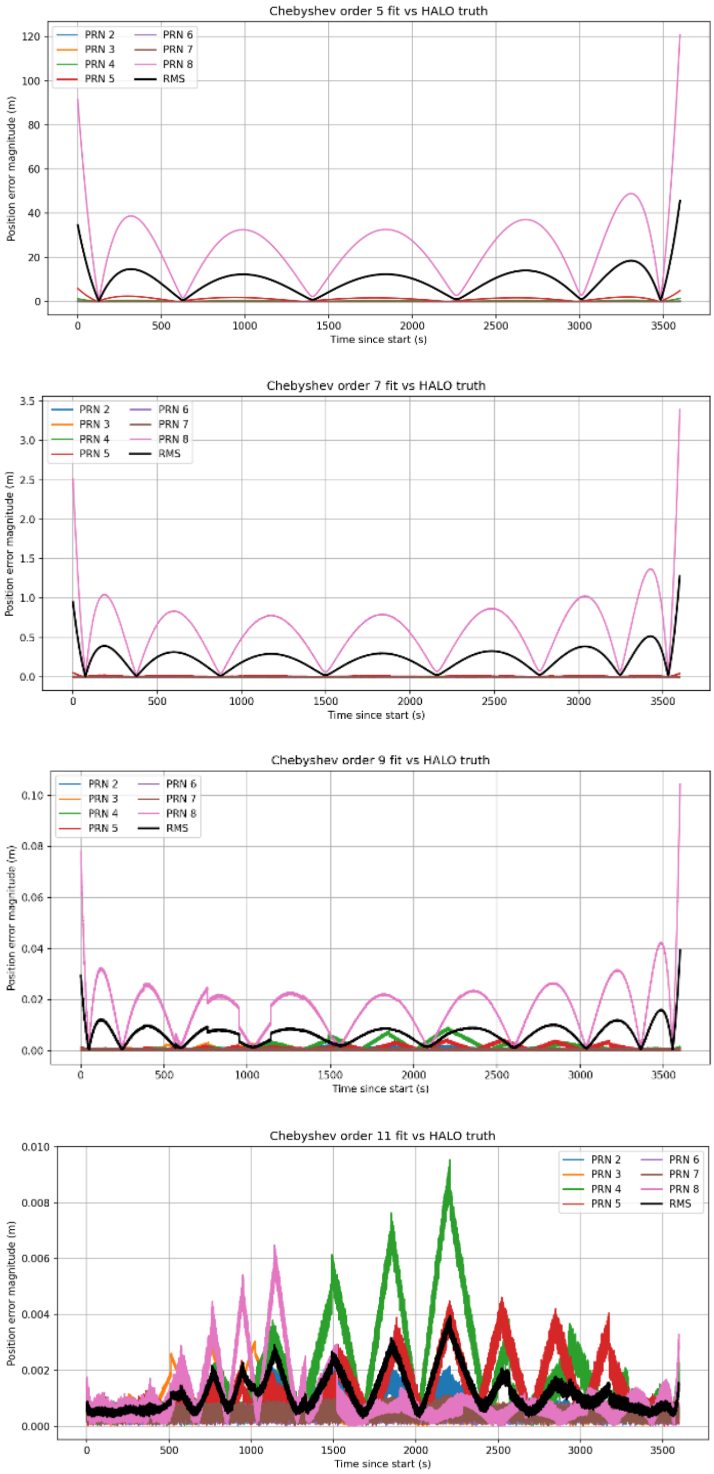

Figure 11 Chebyshev fit trajectories vs. HALO truth (5,7,9,11 degrees)

For comparison, we also implemented a baseline Keplerian + empirical accelerations model with first order and third order harmonics in figure 12, 13, similar to approaches adopted in legacy GNSS. While this method achieves sub-meter performance, residual errors remain at the decimeter level and exhibit slow secular drift within each fit window. The Chebyshev approach therefore offers materially higher fidelity for the highly perturbed ELFO dynamics, particularly when targeting sub-decimeter broadcast orbit accuracy.

These results indicate that compact Chebyshev ephemerides with order 9–11 polynomials and 1-hour segmentation represent an optimal performance, especially on preceision. In particular, an 11-term representation per coordinate meets centimeter class navigation requirements.

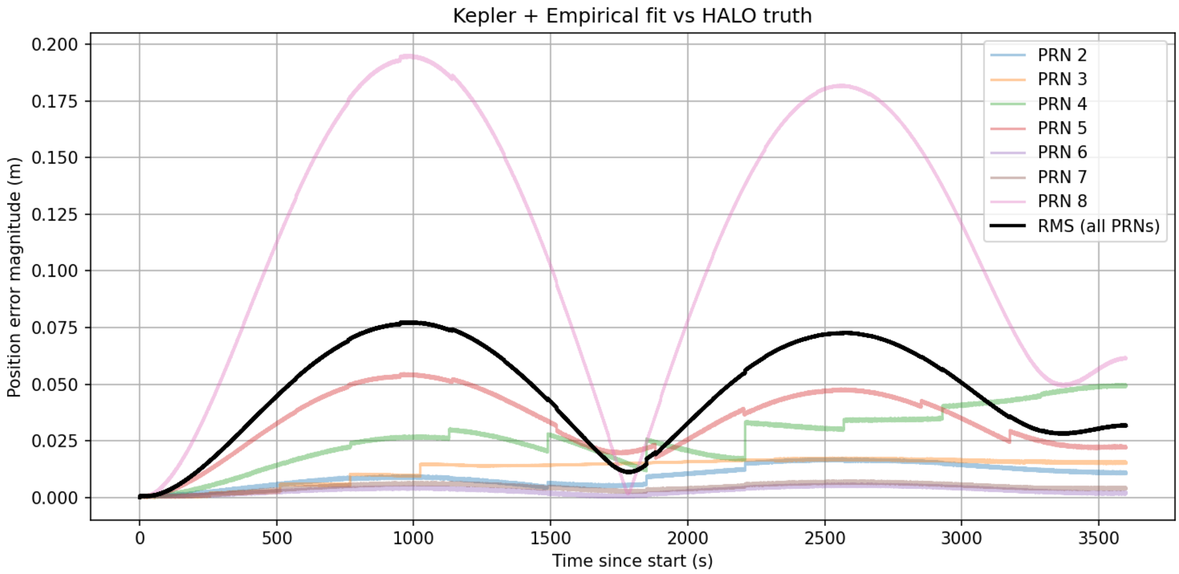

Figure 12 Keplerian empirical fit trajectories vs, HALO truth (1st order harmonic)

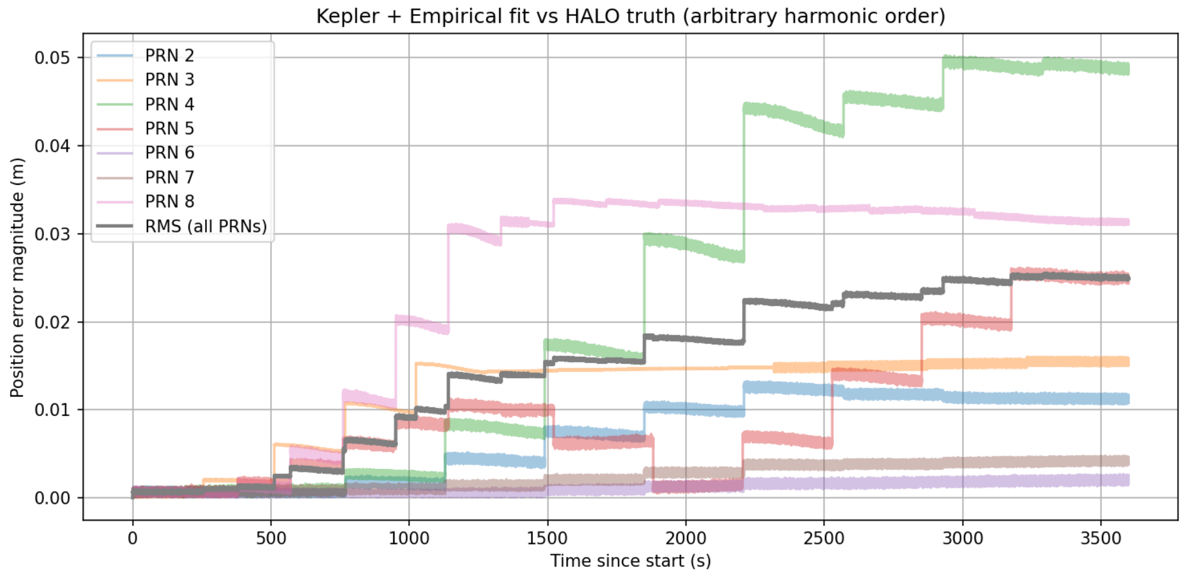

Figure 13 Keplerian empirical fit trajectories vs, HALO truth (3rd order harmonic)

Figure 14 compares the maximum feasible ephemeris validity interval—defined as the longest propagation duration before exceeding the position-error threshold—as a function of true anomaly for both Chebyshev polynomial fits and a Keplerian model augmented with empirical accelerations. In the polynomial case, degree-6 and degree-14 fits demonstrate clear orbital phase dependence: the validity window is shortest near periselene and increases significantly toward aposelene, where orbital motion is slower and the trajectory is locally smoother. The degree-14 model achieves up to ~0.8 orbital periods of valid prediction near the apoapsis region, roughly doubling the performance of the degree-6 case, but both degrade sharply around periselene, where rapid orbital curvature demands higher-order dynamics representation.

A similar trend is observed for the Keplerian + empirical harmonic model. With only the first harmonic term, the fit remains feasible for a fraction of an orbit, with marked degradation near periselene. Introducing the third harmonic significantly improves robustness, enabling validity durations of ~0.6–0.8 orbital periods near apoapsis. It usually has longer maximum interval around the apoapsis. However, even with harmonics, performance near periselene remains constrained.

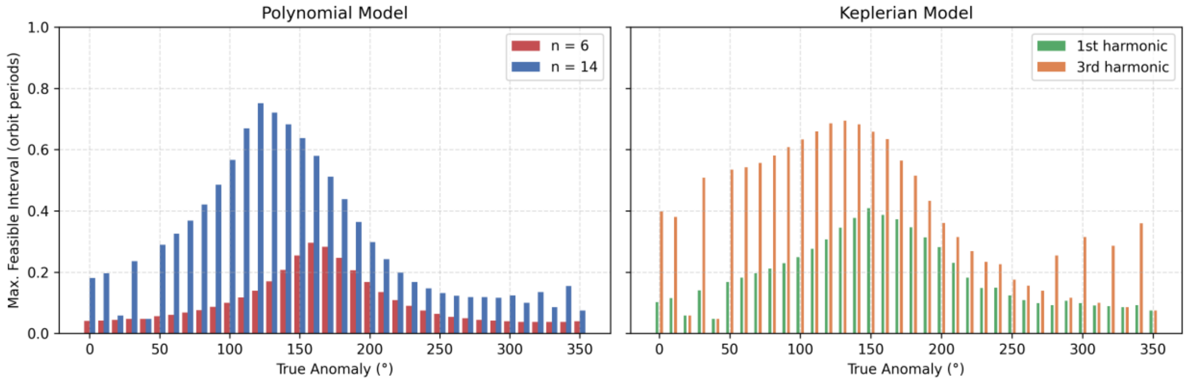

Figure 14 Comparison of feasible approximation intervals for the Chebyshev and Keplerian models using the ELFO

The results highlight a clear performance–cost trade-off between Chebyshev polynomial ephemerides and the Keplerian model with empirical accelerations. High-order Chebyshev fits (e.g., 9–11 terms per axis) achieve superior orbit reconstruction accuracy, consistently outperforming the Keplerian-harmonic method by delivering sub-centimeter to decimeter-level precision across the halo orbit. In contrast, low-order Chebyshev representations do not provide sufficient dynamic fidelity in the highly perturbed lunar environment, and in some cases perform worse than even a first-harmonic Keplerian model.

The empirical-harmonic approach therefore offers a lightweight alternative, achieving moderate accuracy with significantly lower data overhead, particularly at low harmonic order. It is worth mentioning that the Keplerian-based parameterization generally provides longer feasible propagation intervals, because the underlying osculating-element dynamics capture the large-scale orbital motion more robustly than high-order Cartesian polynomials, which tend to lose accuracy rapidly outside their fitting window. However, taking advantage of this benefit requires the receiver to perform onboard numerical orbit propagation, introducing additional computational burden.

Overall, the choice of orbital state parameterization reflects a trade-off between navigation performance and communication and computation cost. High-order Chebyshev polynomials maximize fidelity for precision lunar PNT services, whereas lower-order Keplerian-harmonic models remain efficient for constrained broadcast links or lower-accuracy applications—and additionally offer extended propagation capability that can further reduce the frequency of ephemeris updates.

We will discuss about the exact data bit size to be used in both method in section 7.1.

5.4 Receiver Signal Processing Performance

Figure 15 shows the 2-bit I/Q sample stream produced by the LANS AFS signal simulator. The upper plots display the amplitude histograms of the in-phase (I) and quadrature (Q) outputs after quantization. Both distributions closely resemble Gaussian shapes, which is expected under typical LANS AFS operating conditions where the signal is well below the thermal noise floor. In this low-SNR regime, the received waveform is dominated by noise, causing the I and Q components to exhibit nearly identical statistical behavior. This indicates that the 2-bit quantization scheme still maintains the key noise-like characteristics of the received signal.

Figure 16 depicts the correlation output for one of the Data channels. A well-defined triangular peak is clearly visible, confirming that the simulator-generated signal can be acquired successfully through standard code-correlation processing.

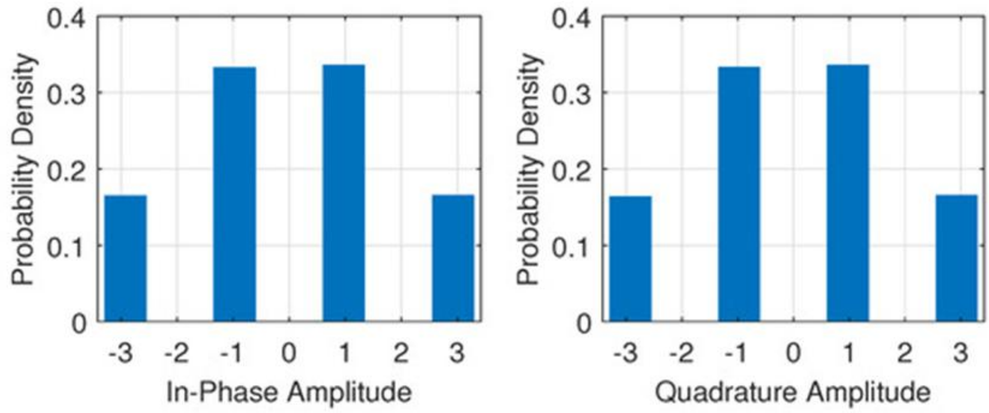

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Figure 15 Histograms of IQ values

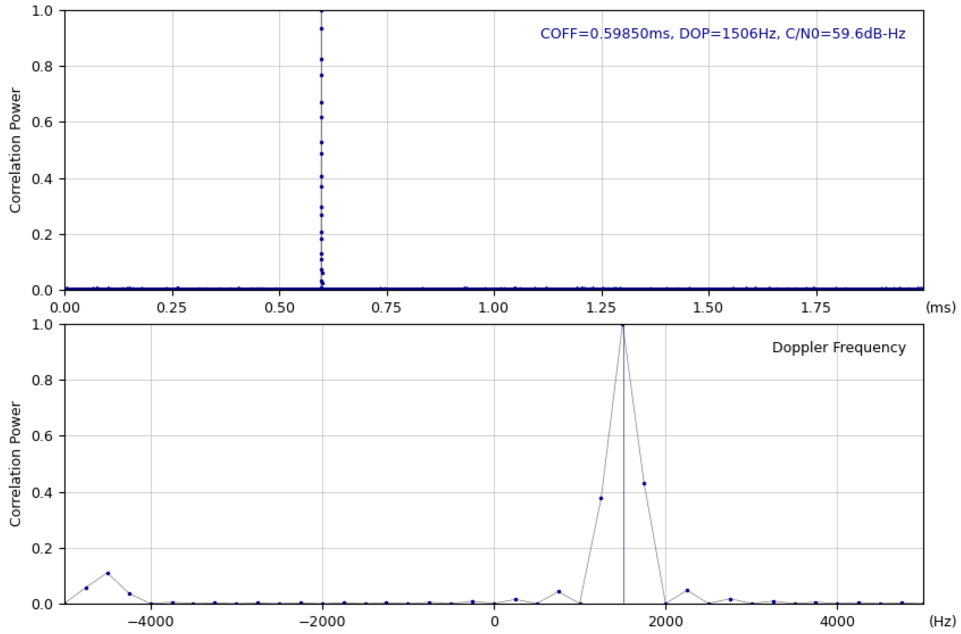


Figure 16 Correlation results for acquisition

The receiver tracking loop performance was evaluated using the DLL, PLL, and correlator output statistics in figure 17, 18. The DLL discriminator output remains zero-mean and bounded, exhibiting only stochastic jitter consistent with the expected thermal noise level and no observable long-term bias or drift, indicating stable code tracking. Similarly, the PLL discriminator output demonstrates tightly clustered phase error with no cycle slips, confirming robust carrier lock and adequate loop bandwidth selection for the received dynamics.

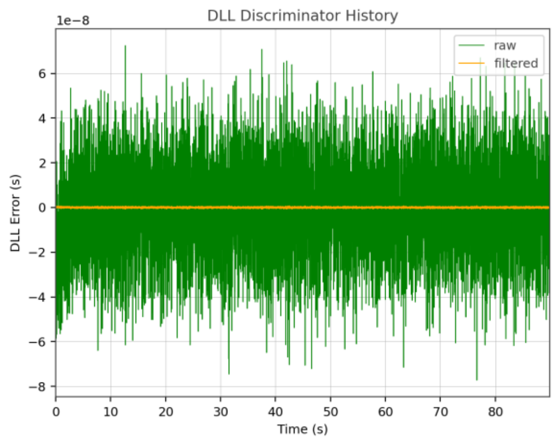


Figure 17 DLL dicriminator raw and filtered out

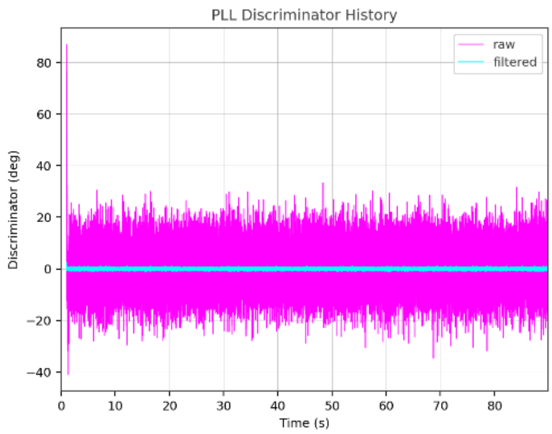


Figure 18 PLL dicriminator raw and filtered out

The prompt and quadrature correlator outputs in figure 19 show well-defined amplitude separation between early/prompt/late channels, and the prompt complex correlation scatter plot in figure 20 forms a compact cluster around the in-phase axis. This behavior is characteristic of a healthy lock condition with high carrier-to-noise ratio, where the residual quadrature component represents thermal noise and minor implementation jitter. The small spread in the IP-QP constellation further validates coherent integration effectiveness and confirms that the code/carrier loops maintain phase alignment throughout the tracking interval.

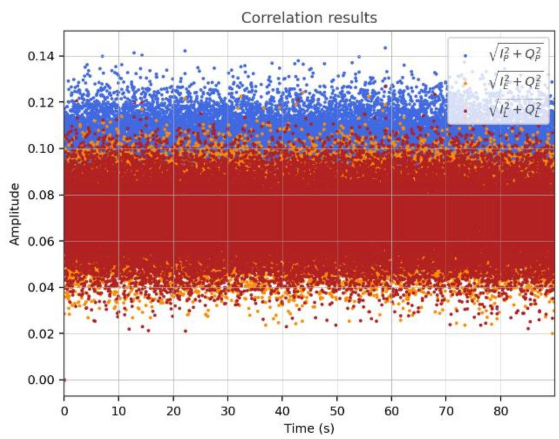


Figure 19 E-P-L correlator results

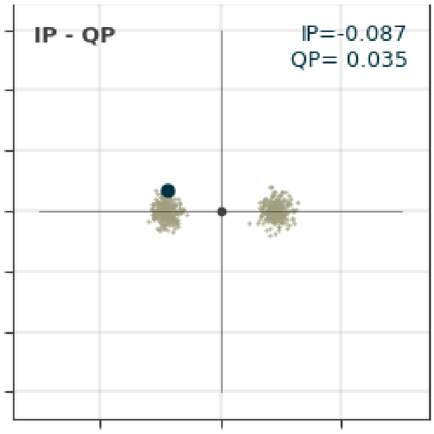


Figure 20 In-phase and quadrature power distribution

5.5 Navigation Performance Analysis

Figures 21 and 22 compare the horizontal positioning performance of the LANS user receiver under different link conditions. In Figure 21, all satellites maintain a carrier-to-noise density ratio (C/N_0) above 35 dB-Hz throughout the observation period, resulting in a stable and well-conditioned geometry. The corresponding North-East error scatter shows a nearly isotropic distribution with a 95 % horizontal error radius of about 0.95 m, indicating sub-meter navigation accuracy.

In contrast, Figure 22 shows degraded signal conditions, where several satellites experience C/N_0 values falling below the threshold or PLL discriminator out above the threshold. If any of the condition happens, the loop will be unlock and has to wait for tracking and relocking. As a consequence, fewer satellites contribute to the navigation solution, leading to an increased Position Dilution of Precision (PDOP) and a less favorable constellation geometry. The resulting horizontal errors exceed 8 m (95 %), forming an elongated distribution that reflects the weakened geometric diversity.

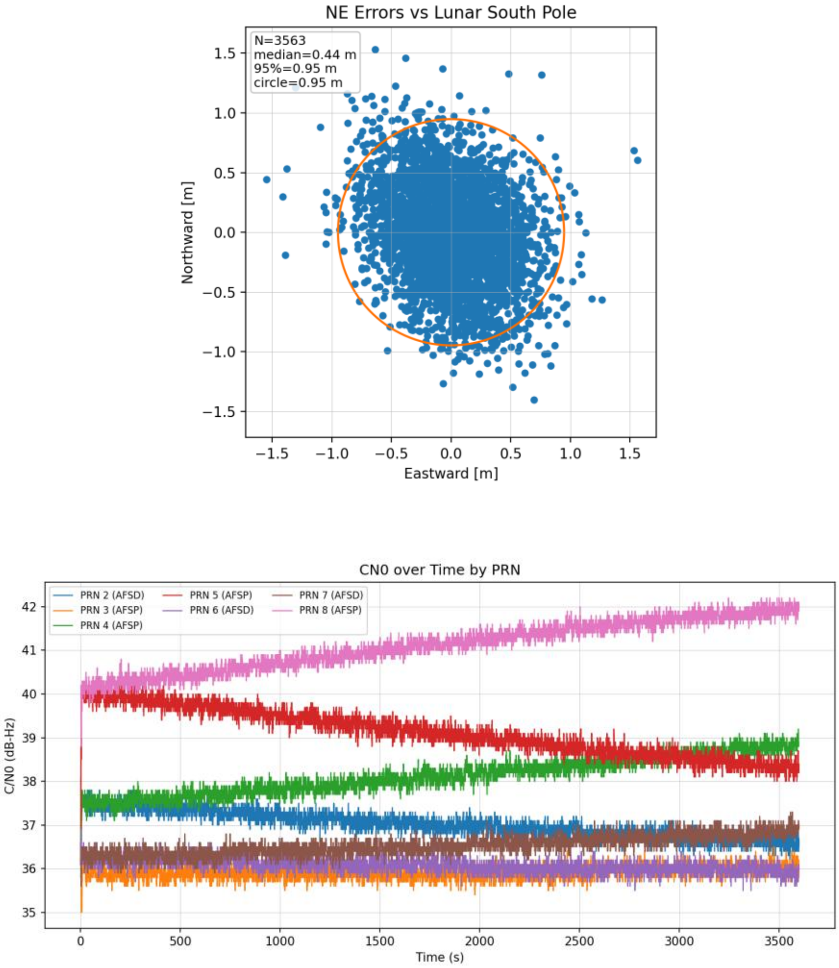
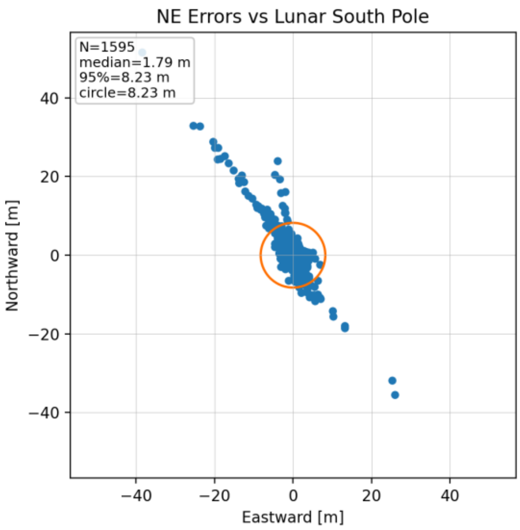


Figure 21 (a) (b) NE error(2 drms) under high C/N_0



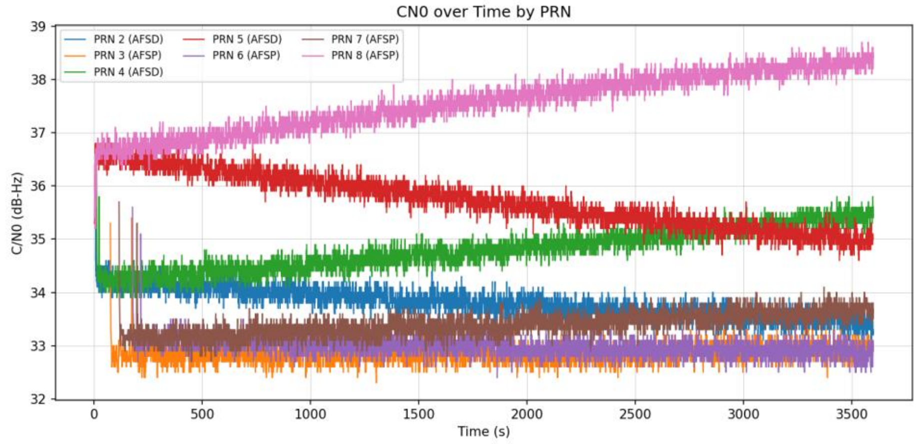


Figure 22 (a) (b) NE error(2 drms) under low C/N_0

For the low C/N_0 condition, a skyplot in figure 23 showing the constellation geometry when there occurs to be the highest PDOP can be analyzed on. As illustrated in Figure XX , at t = 156.9 s, the constellation geometry is highly degraded, yielding a PDOP of 40.9. All visible satellites cluster within a single azimuth sector between 45° and 225°, leaving the north-west sky empty. This coplanar configuration provides limited geometric diversity and almost no vertical leverage, since all satellites are at relatively low elevation angles. Consequently, the line-of-sight vectors become nearly colinear, producing a rank-deficient geometry matrix and a NW-SE amplification of position uncertainty. The resulting error covariance is toward the direction, opposite to the satellite cluster, where no ranging constraint is available. In practical terms, even small pseudorange noise or bias at this epoch would lead to large horizontal deviations and poor altitude observability.

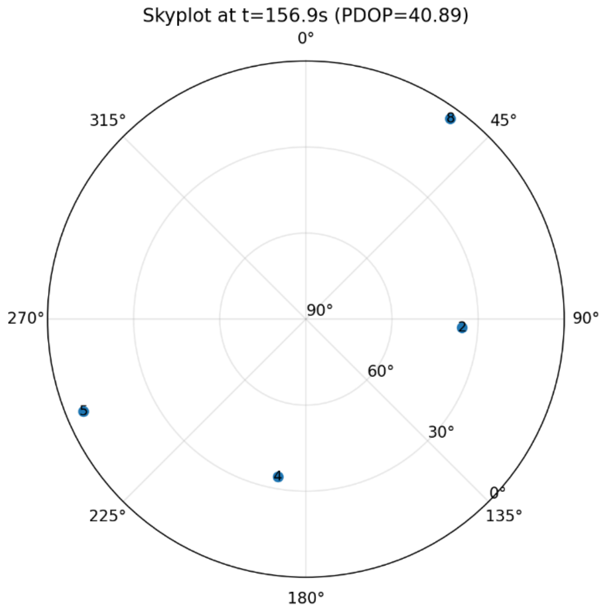


Figure 23 Skyplot over lunar south pole

Chapter 6

6 Error Budget

This section evaluates how different LNSS system design parameters influence the overall navigation error budget. To facilitate this analysis, we establish a baseline scenario in which a single satellite provides navigation to a stationary surface user. The satellite's orbital parameters and initial epoch are listed in Table 2 with PRN=2 and degree of Chebyshev ephemeris is set as 6. The simulation begins at a true anomaly of $f(t_0) = 90°$. The user is positioned at the lunar south pole (LSP) with an elevation mask of $5°$, and no additional surface features or obstructions are considered.

We consider that the navigation satellite maintains knowledge of its orbital and clock states—either through autonomous onboard estimation or ground-based updates. The corresponding clock uncertainties are propagated over time following the methods outlined in Sections 4.4, while the resulting ephemerides are fitted using the procedure described in Section 4.5.2. The satellite is assumed to employ a Rubidium atomic frequency standard (RAFS) [17] or a Micro Rubidium Atomic clock (MIRA) [16] manufactured by Safran Electronics & Defense (2023a) . These choices provide a timing stability comparable to that of terrestrial GNSS systems; the impact of adopting less stable frequency references is analyzed in this section.

For the radiometric link, the satellite transmits the nominal LunaNet Augmented Forward Signal (AFS) centered at 2492.028 MHz, employing a 5.115 Mcps Q-channel spreading code. The signal is radiated with an effective isotropic radiated power (EIRP) of 31 dBW at a maximum off-boresight angle of 15°, consistent with NASA's received surface power specifications (2025). The free-space path loss (FSPL) is modeled explicitly. Multipath is modelled from the experience formula from section 4.3, where (a,b,c) is set as (0.1633, 1.1846, -0.0511). The user terminal is equipped with a hemispherical antenna providing a constant gain of 4 dBi for elevation angles above 5°, and a carrier-aided receiver implementing a first-order delay lock loop (DLL) and a second-order phase lock loop (PLL) to track the pilot Q-channel. The main parameters of this antenna–receiver configuration are summarized in Table 4.

Figure 24 illustrates the variation of the carrier-to-noise density ratio (C/N_0) with elevation angle under the default link conditions. In conventional terrestrial GNSS, satellites in near-circular orbits exhibit C/N_0 growth with elevation, primarily due to reduced atmospheric attenuation and nearly constant FSPL at higher angles. To compensate for low-elevation losses, GNSS satellites often increase antenna gain off-boresight to enhance the EIRP. In contrast, the proposed lunar elliptical orbit experiences no atmospheric attenuation, making FSPL the dominant factor. Consequently, C/N_0 peaks at low elevation angles, where the user–satellite range is shortest. This characteristic can improve link robustness by providing additional margin against multipath, which is typically more severe near the horizon. Alternatively, LNSS designers could implement adaptive transmit-power modulation to maintain a constant received power at the lunar surface, thereby optimizing the satellite's power budget.

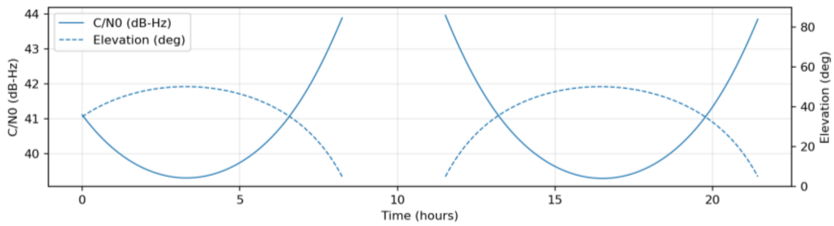


Figure 24 C/N_0 and elevation trend

In this analysis, the navigation satellite updates its ephemeris and clock model at hourly intervals. The user generates pseudorange and time-averaged delta-pseudorange (Doppler) measurements at a rate of one sample per second, whenever the satellite is within view. Figure 25, 26 summarizes the resulting User Equivalent Range Error (UERE) derived from the simulation. The first figure illustrates a representative sample error trajectory and its corresponding 3σ confidence bounds from a single realization, while the second shows the contribution of individual error sources. Each labeled term in the legend represents an error element previously defined in Sections 4.4.

Furthermore, according to the AFS volume-A, the Received Minimum and Maximum Power is assumed to be -160 dbW and -147 dbW so the EIRP would be modified to adapt each power value to evaluate the condition under worst and best C/N_0.

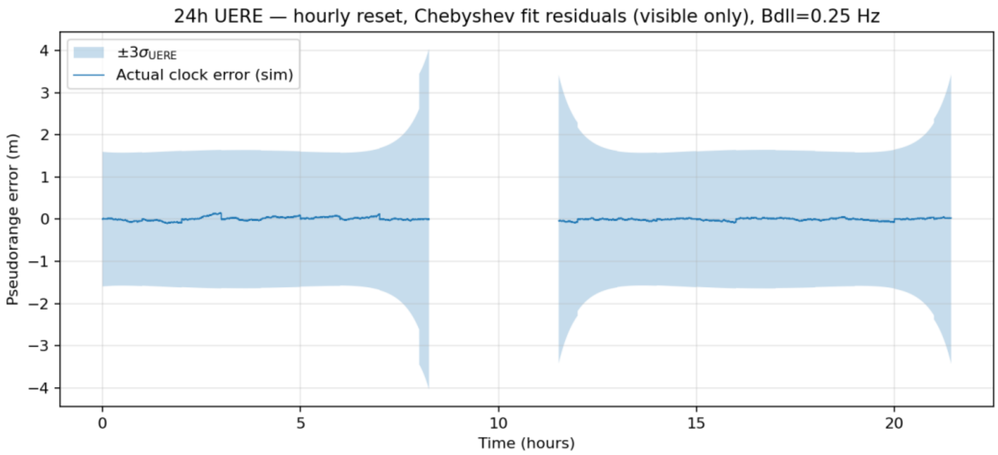


Figure 25 3$\boldsymbol{\sigma}$ pseudorange error using RAFS clock nder received power in -160dbW

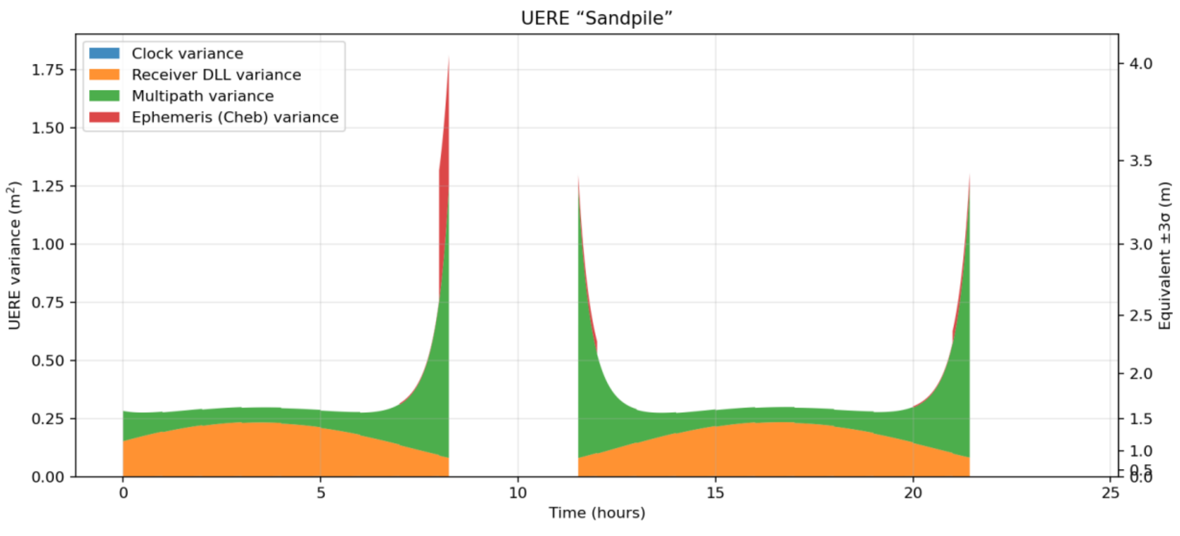


Figure 26 UERE sandpile chart using RAFS clock under received power in -160dbW

As illustrated in the Figure 26, the pseudorange measurement uncertainty is primarily driven by the multipath error ($\sigma_{\rho,path}$) and the Receiver noise errors ($\sigma_{\rho,rec}$). In contrast, the clock noise error is reduced to the centimeter level because of the high frequency stability of RAFS and a short update interval of the clock configuration(1 hour), rendering it negligible relative to other contributions. The ephemeris model error ($\sigma_{\rho,mdl}$) also remains small—on the order of 10^{-5}m—because the update and propagation interval is limited to one hour. Only when the trajectory is over Perilune then the ephemeris error becomes obvious because of the greater dynamics and velocity here.

Notably, the sample error trajectories of clock error exhibit temporal correlation rather than pure white-noise behavior. Although $\sigma_{\rho,clk}$ originate as random perturbations in the true state, these errors evolve according to the system dynamics, resulting in gradual, exponentially growing trends over time.

In comparison, when the received power increases to around –147 dBW (Figure 27, 28), the overall UERE decreases significantly across all components due to the higher C/N_0. The improvement is most evident in the receiver noise term ($\sigma_{\rho,rec}$), which benefits directly from the enhanced signal strength, resulting in sub-centimeter-level contributions. The multipath component ($\sigma_{\rho,path}$) remains largely unchanged, as it depends more on surface geometry than on received power.

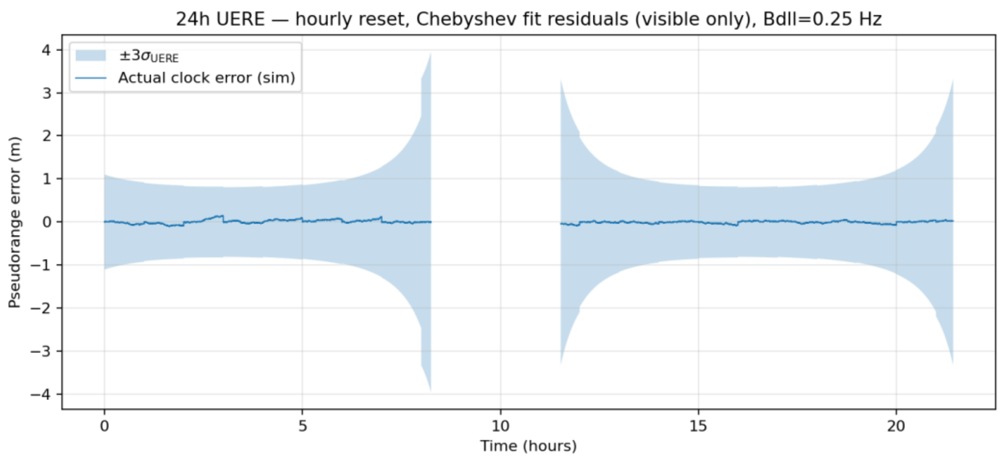


Figure 27 3$\boldsymbol{\sigma}$ pseudorange error using RAFS clock nder received power in -147dbW

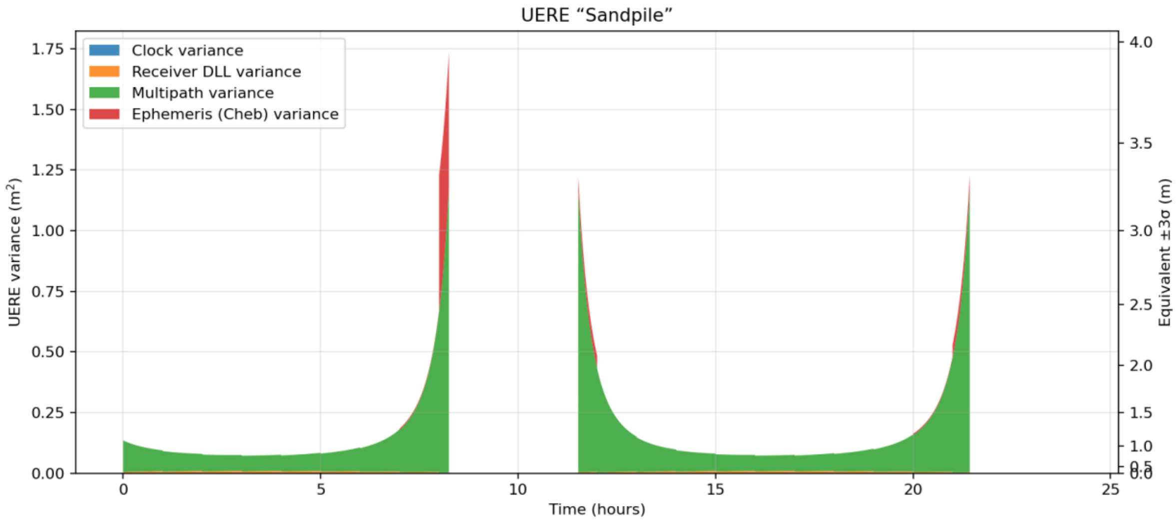


Figure 28 UERE sandpile chart using RAFS clock under received power in -147dbW

In the baseline simulation presented above, both the satellite and user were equipped with clocks providing excellent short-term stability, albeit with high size, weight, and power (SWaP) requirements. For comparison, the simulation was repeated using clocks with lower SWaP but reduced stability. In this configuration, the satellite employs a Safran MIRA.As summarized in Table 5, these devices represent order-of-magnitude reductions in both SWaP and timing stability.

Figure 29, 30 illustrates the impact of this clock substitution. The sandpile plots show that the satellite's clock-model error and short-term instability become dominant contributors to the total uncertainty, since the MIRA exhibits poorer short- and long-term performance than the RAFS used in the baseline case. Overall, replacing the high-stability RAFS with lower-grade oscillators increases uncertainty immediately after each coefficient update and causes the total uncertainty to diverge more rapidly with time.

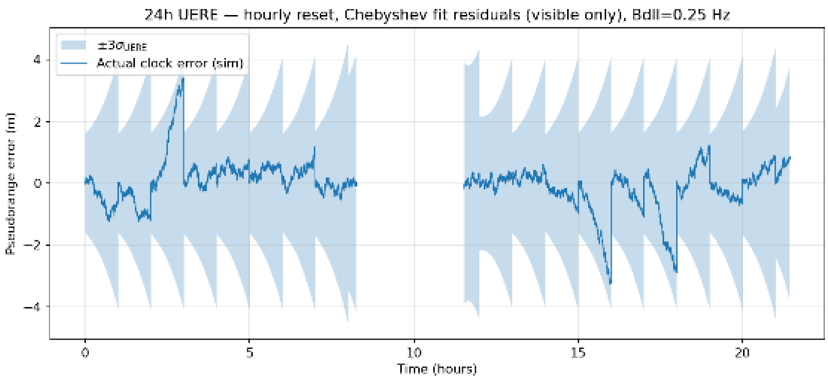


Figure 29 3$\boldsymbol{\sigma}$ pseudorange error using MIRA clock under received power in -160dbW

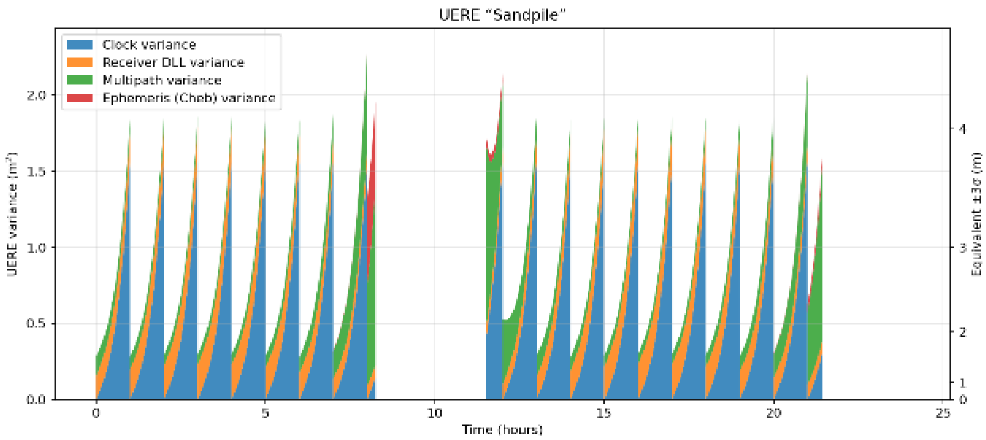


Figure 30 UERE sandpile chart using MIRA clock under received power in -160dbW

Antenna/Link Parameters		Tracking Loop Parameters	
elevation mask	5°	Front-end bandwidth	10.23 MHz
Gain	4 dBi	Code noise bandwidth	0.25 Hz

Antenna/Link Parameters		Tracking Loop Parameters	
Noise temperature	150 K	Carrier noise bandwidth	15 Hz
Noise figure	–5 dB	Code integration time	10 ms
Conversion losses	–1.5 dB	Carrier integration time	10 ms
System losses	–2 dB	E/L correlator spacing	0.25 chips

Table 4 Antenna/link parameters& tracking loop parameters

Clock	Size (cm^3)	Weight (g)	$\sigma_y(\tau = 10s)$ (s/s)	$\sigma_y(\tau = 10000s)$ (s/s)
Safran RAFS	41.3	3400	1.3e-11	5e-12
Safran mRO-50	3148.6	94	1e-12	3e-14

Table 5 RAFS and MIRA configuration [16,17]

Chapter 7

7 Discussion

This project successfully demonstrated an end-to-end software-defined simulation of the Lunar Augmented Navigation Service (LANS). The results validate the feasibility of using the Augmented Forward Signal (AFS) for positioning at the lunar south pole. However, several critical design trade-offs and limitations in the current simulation model warrant further discussion.

7.1 Ephemeris Parameterization Trade-offs

To compare the broadcast efficiency of different orbit parameterization schemes, we estimate the number of bits required to encode all parameters of a single validity segment. Two models are evaluated: (1) a Cartesian polynomial (Chebyshev-like) model, and (2) a Kepler + empirical acceleration model.

a. Polynomial model

Each Cartesian component (x, y, z) is represented by an independent degree-$\boldsymbol{n}$ polynomial, giving $3(n+1)$ coefficients. Two additional metadata fields (reference epoch $\boldsymbol{t_0}$and interval duration Δt) are included. Following common practice in Chebyshev-based orbit compression studies, every polynomial coefficient is quantized with 17 bits, while metadata fields use 16 bits. The resulting bit budget is:

$$B_{\text{poly}}(n) = 3(n+1)\, b_{\text{coeff}} + 2\, b_{\text{meta}}, b_{\text{coeff}} = 17,\, b_{\text{meta}} = 16.$$

b. Kepler + empirical RSW model

The dynamical model consists of seven osculating Kepler parameters $(a, e, i, \Omega, \omega, M_0, n)$ and one reference epoch, giving eight real-valued parameters. In line with GPS LNAV practice, the seven Keplerian parameters are assigned 32 bits each, and the epoch field 16 bits. Perturbations are captured using empirical accelerations in the RSW frame:

$$a_u(t) = c_{u,0} + \sum_{k=1}^{n_{\text{kep}}} (c_{u,cos\,,k} cos\,(knt) + c_{u,sin\,,k} sin\,(knt)), u \in \{R, S, W\}.$$

This yields $3(1 + 2n_{\text{kep}})$empirical coefficients, each quantized using 16 bits. The total bit budget is:

$$B_{\text{kep+emp}}(n_{\text{kep}}) = (7b_{\text{kep}} + b_{\text{epoch}}) + 3(1 + 2n_{\text{kep}})b_{\text{emp}},$$

with $b_{\text{kep}} = 32,\ b_{\text{epoch}} = 16,\ b_{\text{emp}} = 16.$

These quantization rules provide a framework for comparing the broadcast data requirements of Keplerian and polynomial orbit representations. Table 6 indicates Model Parameter and Data Size Comparison between different orders and models.

Model	Params	Bits	Bytes
Polynomial (n = 5)	20	338	42.2 B
Polynomial (n = 7)	26	440	55.0 B
Polynomial (n = 9)	32	542	67.8 B
Polynomial (n = 11)	38	644	80.5 B
Kepler + RSW empirical (n_{kep} = 1)	17	384	48.0 B
Kepler + RSW empirical (n_{kep} = 3)	29	576	72.0 B

Table 6 Model Parameter and Data Size Comparison

Looking back to section 5.3 , we can make a conclusion that high-order Chebyshev polynomials provide excellent local approximation accuracy, and their reconstruction error decreases rapidly as the polynomial degree increases. However, this gain comes at the cost of a proportional increase in the number of coefficients—and therefore the broadcast bit budget—making high-order Chebyshev models comparatively expensive in terms of data volume. In contrast, the Kepler + empirical model achieves reasonable accuracy with only a few low-order harmonic terms, offering a very compact representation. Increasing the harmonic order beyond a modest level produces only marginal improvements, while the parameter count and computational burden grow significantly, and the approach requires on-board numerical integration during reconstruction. Nonetheless, Kepler-based models retain an important advantage: because they encode the underlying dynamics rather than only a finite-interval approximation, they maintain useful predictive accuracy outside the fitted window, whereas Chebyshev segments degrade rapidly once extrapolated. Overall, the choice between the two models reflects a fundamental trade-off between local accuracy vs. broadcast size, and between dynamic interpretability (Kepler) and purely numerical precision (Chebyshev).

7.2 Unmodelled Error Sources and UERRE

While this simulation incorporates key environmental factors such as free-space path loss and elevation-dependent multipath, several error sources remain unmodelled and represent limitations of the current study:

a.Receiver Dynamics and Oscillator Stress:

The simulation placed the user at a static location (Shackleton Crater). In a dynamic scenario, such as a rover traversing rough terrain, the receiver would experience dynamic stress on the tracking loops. Vibration-induced phase noise in the local oscillator and rapid changes in Doppler shift could degrade the Phase Lock Loop (PLL) performance, potentially causing cycle slips or loss of carrier lock that are not observed in the static test.

b. ODTS modelling

Most GNSS satellites in Earth orbit operate in high-altitude, nearly circular trajectories where the dynamical environment is relatively benign and long-term perturbations are small. These spacecraft typically carry high-stability Rubidium or even more advanced onboard clocks (Dupuis et al., 2008), and they are continuously monitored by a dense, globally distributed ground network. The combination of stable orbital motion, frequent tracking updates, and precise timing hardware ensures that both orbital ephemeris and clock errors remain well-bounded over time.

By contrast, navigation satellites deployed in cislunar space experience a far more complex dynamical regime. Pronounced third-body perturbations from the Earth and Sun, together with the Moon's highly irregular gravity field, cause rapid divergence in propagated states. At the same time, long-range tracking from Earth becomes intermittent and geometry-limited, reducing the ability of ground networks to provide frequent orbit and clock corrections. Consequently, a Lunar Navigation Satellite System (LNSS) may need to rely much more heavily on onboard, autonomous real-time state estimation, typically implemented through some form of sequential filtering (e.g., EKF/UKF). Therefore, such error like orbit Propagation Error and time synchronization error can be modelled.

1. Orbit Propagation Error

- Definition: It is the error caused by the discrepancy between the satellite's true trajectory and the predicted trajectory used by the satellite to generate navigation data.
- Cause: It arises from imperfections in the dynamic models used to predict the orbit, particularly due to complex perturbations in the lunar environment (e.g., third-body effects from Earth/Sun and lunar gravity harmonics).

2. Time Synchronization Error (Satellite Clock Error)

- Definition: It is the error caused by the discrepancy between the satellite's true clock offset (relative to the system time scale, e.g., GPST or Lunar System Time) and the estimated or predicted clock parameters (bias, drift, and drift rate) broadcast to the user.
- Cause: It arises from a combination of physical hardware instability and modeling limitations such as Inherent Clock Instability, Relativistic Modeling Errors and ODTS Estimation Noise.

c. User Equivalent Range-Rate Error (UERRE):

The current analysis focused primarily on position domain errors (UERE). However, velocity estimation is critical for lunar orbit insertion and landing. The UERRE depends heavily on the short-term stability (Allan deviation) of the clocks and the thermal noise in the PLL. While Doppler measurements were generated, a comprehensive velocity accuracy assessment accounting for receiver dynamics and jerk was not performed. Future work could quantify UERRE to validate the system's capability for precision landing operations.

Chapter 8

8 Conclusion

This thesis has developed and validated an end-to-end, software-defined radio framework for lunar navigation using the Lunar Augmented Navigation Service (LANS) Augmented Forward Signal (AFS). Starting from the LunaNet and LANS concepts, the work formulated lunar pseudorange/Doppler observation models, defined a consistent set of lunar frames and time assumptions, and constructed a full PNT error budget tailored to the lunar south-pole environment. On this foundation, a complete SDR-based signal chain was implemented, from orbit and ephemeris generation through AFS waveform synthesis, channel effects, receiver acquisition/tracking/decoding, and least-squares PVT, enabling realistic evaluation of lunar user performance in the absence of any real AFS broadcasts.

Using high-fidelity HALO-derived ELFO trajectories, two broadcast ephemeris parameterization methods—Keplerian elements with RSW empirical accelerations and Chebyshev Cartesian polynomials—were implemented and compared in terms of accuracy, bit cost, and user-side computational burden. The simulator and receiver were then used to evaluate an ELFO constellation over the Shackleton region, quantifying coverage, PDOP, UERE/UERRE, and resulting 3D position errors. The results show that the baseline two-plane ELFO geometry can provide continuous single- and dual-satellite visibility and high four-satellite availability over the south pole, with meter-level positioning performance achievable under realistic link budgets and clock assumptions. The ephemeris trade-off analysis confirms that Chebyshev representations can deliver higher accuracy at the cost of additional bits and limited extrapolation, while Kepler+empirical models remain more bit-efficient but shift complexity to on-board integration and user receivers.

Overall, the outcomes of this work meet the stated objectives. A flexible, SDR-based AFS simulator and a compatible reference receiver have been realized; a lunar-specific observation and error model has been constructed; and the combined framework has been used to assess

ELFO constellation performance and ephemeris design choices for a representative south-pole users.

Future research should aim to construct a more realistic lunar-surface scenario by incorporating receiver dynamics, lunar rotation, and a more comprehensive channel model that captures terrain-induced multipath, scattering, and other propagation effects. Further developments can also expand pilot-assisted tracking to improve support for low-SNR operations and introduce lunar-specific navigation solvers with multipath-mitigation capabilities, enabling more robust performance in shadowed or cluttered regions near the poles. In addition, hardware-in-the-loop experiments could be carried out to further enhance realism and validate the end-to-end behaviour of the proposed system under practical SDR front-end constraints.

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Appendix

Appendix A — Keplerian Orbit Element to State Vector Conversion

This appendix summarises the standard two-body orbital mechanics formulas used to convert classical Keplerian orbital elements into Earth–Moon inertial (ECI/MCI) position and velocity vectors. The equations listed here are used throughout the thesis to generate reference trajectories, validate orbital dynamics models, and provide initial conditions for the AFS signal simulator.

A.1 Constants

Constant	Value / Description
π	3.1415926535898
μ	$4.9027890 \times 10^{12}\ \mathrm{m^3/s^2}$ (geocentric gravitational constant of the Moon)

A.2 Orbital Geometry and Anomaly Relations

The following expressions describe the geometric relationships for a Keplerian orbit.

Semi-latus rectum

$$p = a(1 - e^2)$$

Mean motion

$$n = \sqrt{\frac{\mu}{a^3}}$$

Mean anomaly

$$M = n(t - T_0)$$

Kepler's Equation (solve for eccentric anomaly E)

$$M = E - e\sin E$$

True anomaly

$$v = \tan^{-1}\left(\frac{\sqrt{1-e^2}\sin E}{\cos E - e}\right)$$

Orbital radius

$$r = a(1 - e\cos E)$$

A.3 Orbit-Plane Basis Vectors

The orientation of the orbital plane relative to the inertial frame is determined by the right ascension of the ascending node Ω, inclination i, and argument of perigee ω.

P vector

$$P = \begin{bmatrix} \cos\Omega\cos\omega - \sin\Omega\cos i\sin\omega \\ \sin\Omega\cos\omega + \cos\Omega\cos i\sin\omega \\ \sin i\sin\omega \end{bmatrix}$$

Q vector

$$Q = \begin{bmatrix} -\cos\Omega\sin\omega - \sin\Omega\cos i\cos\omega \\ -\sin\Omega\sin\omega + \cos\Omega\cos i\cos\omega \\ \sin i\cos\omega \end{bmatrix}$$

A.4 Position Vector in Inertial Coordinates

The position of the satellite in the inertial frame is obtained by combining the radial distance $r$and true anomaly $v$with the basis vectors:

$$X = r\cos v \cdot P + r\sin v \cdot Q$$

A.5 Velocity Vector in Inertial Coordinates

The corresponding velocity vector is:

$$\dot{X} = \sqrt{\frac{\mu}{p}}\sin v \cdot P + \sqrt{\frac{\mu}{p}}(e + \cos v) \cdot Q$$

This formulation follows from the two-body energy integral and the geometry of orbital motion.

Appendix B — User Algorithm for Deriving the Acceleration Vector in the Satellite Inertial Frame

This appendix summarises the user-side algorithm required to reconstruct the empirical accelerations broadcast in the LANS ephemeris message. Given the satellite position and velocity at a given epoch, the algorithm computes the radial (R_u), along-track (W_u), and cross-track (S_u) unit vectors, evaluates the empirical acceleration components, and transforms them into the inertial reference frame.

B.1 Computation Procedure

Computation	Description
$u_0 = v_0 + \omega$	Argument of latitude at the initial epoch
$\Delta u = u_0 + \sqrt{\frac{\mu}{a^3}}(t - T_0) \cdot 2\pi$	Argument of latitude increment at time t(seconds of day)
$r = \parallel X \parallel$	Magnitude of satellite position vector
$R_u = \frac{X}{r}$	Radial unit vector
$W_u = \frac{X \times \dot{X}}{\parallel X \times \dot{X} \parallel}$	Along-track (cross-product) unit vector
$S_u = R_u \times W_u$	Cross-track unit vector
$R = R_0 + R_S \sin \Delta u + R_C \cos \Delta u$	Radial empirical acceleration
$W = W_0 + W_S \sin \Delta u + W_C \cos \Delta u$	Along-track empirical acceleration
$S = S_0 + S_S \sin \Delta u + S_C \cos \Delta u$	Cross-track empirical acceleration

Computation	Description
$a = RR_u + WW_u + SS_u$	Acceleration vector expressed in the inertial frame

B.2 Algorithm Explanation

1. Argument of Latitude

   The initial argument of latitude u_0 combines the true anomaly v_0 and argument of perigee ω.

   The increment Δu increases linearly with mean motion, enabling the empirical acceleration components to vary with orbital phase.

2. Construction of the RSW Frame

   - R_u: radial direction from the central body toward the spacecraft
   - W_u: direction of orbital angular momentum (along-track)
   - S_u: completes the right-handed RSW triad

3. Empirical Acceleration Reconstruction

   The broadcast ephemeris includes harmonic coefficients $(R_0, R_S, R_C, \ldots)$.

   These reconstruct periodic accelerations that compensate for unmodelled perturbations (e.g., lunar gravity harmonics, SRP).

4. Transformation to Inertial Frame

   After computing R, W, S, the acceleration is expressed in the inertial frame via a linear combination of the three unit vectors.

B.3 Final Expression

$$\boxed{a(t) = (R_0 + R_S \sin \Delta u + R_C \cos \Delta u)R_u + (W_0 + W_S \sin \Delta u + W_C \cos \Delta u)W_u + (S_0 + S_S \sin \Delta u + S_C \cos \Delta u)S_u}$$

This expression yields the total acceleration acting on the spacecraft, consistent with the broadcast empirical model used in the AFS/LCNS ephemeris.